

LEFT VENTRICULAR HEMODYNAMICS, HEART FAILURE, SHOCK, AND MECHANICAL CIRCULATORY SUPPORT

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In 1856, Carl Ludwig first described the intricate interdependence of end-diastolic volume and cardiac stroke volume: "A strong heart that is filled with blood empties itself more or less completely, in other words, changes the extent of the contractile power." Numerous other scientists and physicians passing through the laboratory of Carl Ludwig advanced the understanding of ventricular hemodynamics until Otto Frank, who studied with Ludwig from 1891 to 1893, defined the relationship of ventricular volume and pressure based on an understanding of the properties of skeletal muscle and described this in the form of a pressure-volume loop. Frank's interpretation that cardiac muscle function is based on a simple relationship between length and tension proved to be minimalistic. Ernest Henry Starling, in the early 1900s, working primarily with mammalian hearts, added complexity to this idea and published a series of lectures concluding "the rise of venous pressure must be regarded as one of the mechanical means which are operative in enabling the heart to maintain an output corresponding to the blood it receives from the venous system." Starling noted, however, that many other components are at play in determining the final stroke volume of the heart. Starling also put forth the idea that clinical heart failure occurs when increases in preload, and thus increases in contractility, can no longer compensate for the failing heart muscle. The principles governing the interplay between end-diastolic volume, afterload, contractility, and stroke volume became known as *Starling's Law of the Heart* and are represented by the Frank-Starling curve. And so was born the concept of ventricular hemodynamics. It was not until the 1960s that the concept of left ventricular pressure-volume relationships was revived, and a better understanding of left ventricular mechanics as displayed by the current pressure-volume loop emerged.^{1,2}

LEFT VENTRICULAR HEMODYNAMICS

To understand the pathophysiology of the various etiologies of heart failure, cardiogenic shock, and the therapies with which they can be treated, it is essential to have a working knowledge of left ventricular hemodynamics. This begins with a thorough understanding of the cardiac cycle (Fig. 11.1). The cardiac cycle consists of two phases: systole and diastole. Left ventricular systole begins when the pressure in the atrium falls below that of the ventricle and the mitral valve closes. A period of isovolumic contraction occurs during which the pressure in the ventricle builds while the volume remains constant until ventricular pressure exceeds aortic pressure, opening the aortic valve. Opening of the aortic valve begins the second phase of systole, ejection, during which systolic pressure continues to rise before falling with the ejection of ventricular volume. Once the left ventricular pressure falls below aortic pressure, the aortic valve closes and diastole begins. As with systole, diastole begins with an isovolumic period during which the ventricle relaxes without change in volume. It is important to note that this relaxation is active and requires energy. Once the pressure in the ventricle falls below that of the atrium, the mitral valve opens and the filling phase of diastole begins. Ventricular filling consists of two phases: passive filling and filling secondary to atrial contraction. Under normal circumstances, the pressure in a normal ventricle rises little in response to increased volume because of continued active relaxation. The cycle completes once the pressure in the ventricle exceeds that in the atrium and the mitral valve closes.

There are numerous factors that contribute to myocardial performance, including afterload, preload, heart rate, myocardial contractility, diastolic parameters, and microvascular function. There are many graphical ways that the interactions between these parameters and the effects on ventricular hemodynamics can be displayed. The Starling curve and the force-tension curve are two easy mechanisms to display the interaction between preload, afterload, contractility, and stroke volume (Fig. 11.2). The Starling curve displays the interaction between the preload and stroke volume. As ventricular volume increases, leading to increased ventricular stretch and active tension, the stroke volume increases if other parameters remain constant. An upward shift in the Starling curve, signifying an increased stroke volume per amount of preload, results from either an increase in contractility or a decrease in afterload. Conversely, a downward shift in the Starling curve signifies a decreased stroke volume per amount of preload and results from either a decrease in contractility or an increase in afterload (see Fig. 11.2A). The force-tension curve displays the interaction between the afterload and stroke volume. As afterload increases, the stroke volume falls if other parameters remain constant. An upward shift in the force-tension curve, signifying an increase in the stroke volume at a constant afterload, results from either an increase in contractility or an increase in preload. Conversely, a downward shift in the force-tension curve signifies a decrease in the stroke volume at a consistent afterload and results from either a decrease in

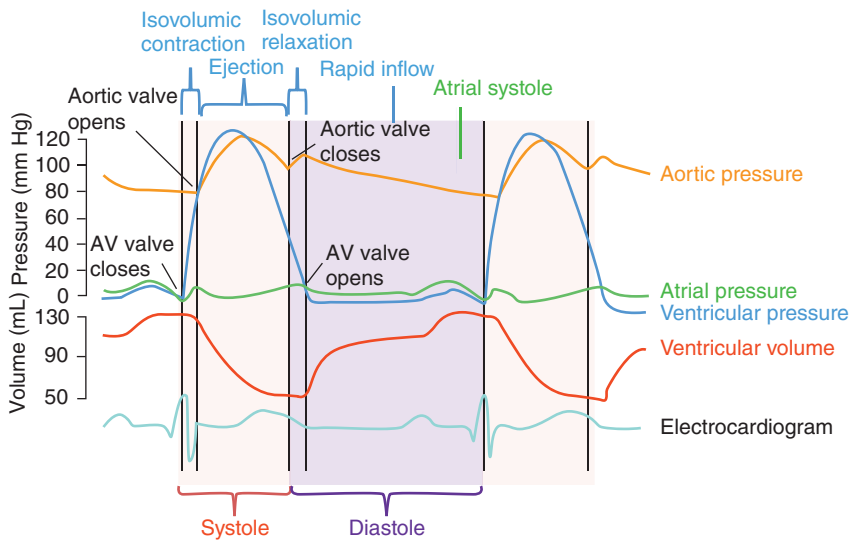


Fig. 11.1 The cardiac cycle. A single cardiac cycle consists of systole and diastole. Systole is composed of isovolumic contraction and ejection. Diastole is composed of isovolumic relaxation and filling. This diagram demonstrates the interactions between atrial pressure, ventricular pressure, aortic pressure, and ventricular volume. AV, Atrioventricular.

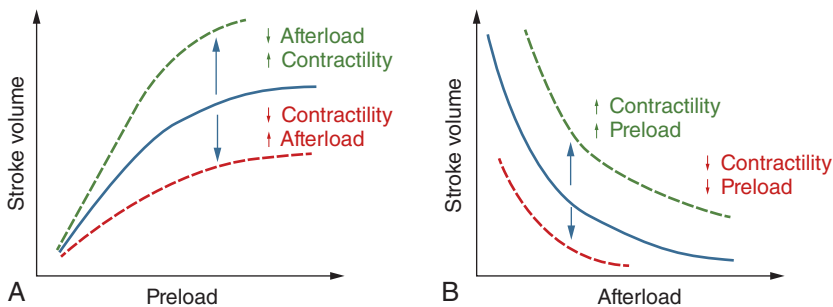


Fig. 11.2 (A) Starling curve. The Starling curve displays the effects of preload on the stroke volume. At a constant preload, an upward shift in the curve (denoted by the upward blue arrow and the green line) signifies either an increase in contractility or a decrease in afterload. A downward shift (denoted by the downward blue arrow and the red line) represents a decrease in contractility or an increase in afterload. (B) Force-tension curve. The force-tension curve displays the effects of afterload on the stroke volume. At a constant afterload an upward shift of the curve (denoted by the upward blue arrow and the green line) represents either an increase in contractility or preload. A downward shift (denoted by the downward blue arrow and the red line) represents a decrease in contractility or preload.

contractility or a decrease in preload (see Fig. 11.2B). These two curves in combination can be used to describe the effects of certain actions on ventricular hemodynamics. For example, in Fig. 11.3, moving from point A to point B on the Starling curve demonstrates a downward shift and a decrease in the stroke volume at a constant preload. This signifies either a decrease in contractility or an increase in afterload. Moving from point A to point B on the force-tension curve clearly demonstrates that the decrease in the stroke volume is due to an increase in afterload. Both contractility and preload are unchanged, as the original force-tension curve has not shifted up or down.

THE PRESSURE-VOLUME LOOP

The information in the cardiac cycle diagram, the Starling curve, and the force-tension curve can be depicted in a single illustration: the pressure-volume loop. This is essentially a plot of the left ventricular pressure versus volume at

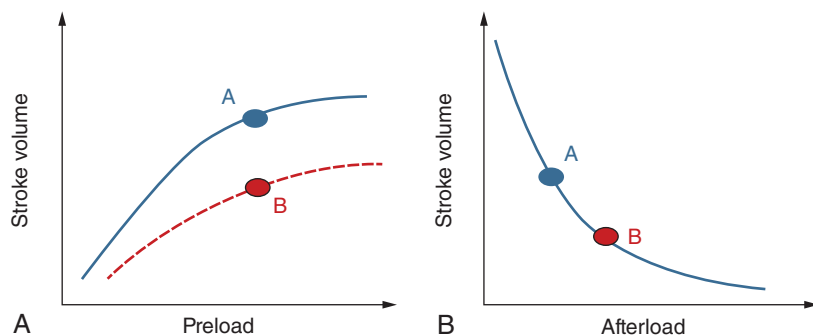


Fig. 11.3 Using the Starling and force-tension curves together to determine the hemodynamic effects. (A) Moving from point A to point B on the Starling curve signifies either a decrease in contractility or an increase in afterload. (B) The force-tension curve demonstrates that moving from point A to point B is an increase in afterload.

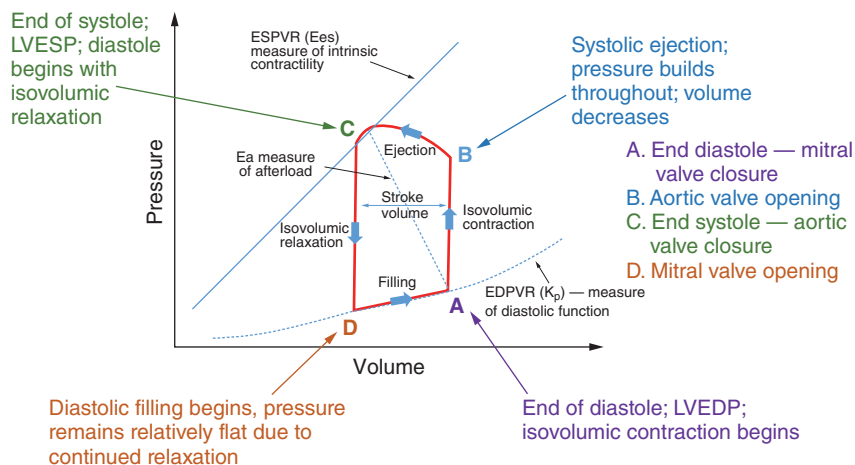


Fig. 11.4 The pressure-volume loop. The pressure-volume loop is a way of displaying the cardiac cycle and several hemodynamic characteristics of the left ventricle. *Ea*, Arterial elastance; *EDPVR*, end-diastolic pressure-volume relationship; *Ees*, end-systolic elastance; *ESPVR*, end-systolic pressure-volume relationship; *K_p*, modulus of chamber stiffness; *LVEDP*, left ventricular end-diastolic pressure; *LVES*, left ventricular end-systolic pressure.

numerous time points during the cardiac cycle (Fig. 11.4). A basic understanding of the pressure-volume loop can be immensely helpful in understanding the pathophysiology of left ventricular failure, how therapies may alter ventricular hemodynamics, and whether those therapies will be helpful or harmful for certain conditions. There is an enormous amount of information housed within the pressure-volume loop. First, the pressure-volume loop is simply a diagrammatic way of illustrating the cardiac cycle. Point A signifies the end of diastole, when ventricular pressure exceeds atrial pressure, closing the mitral valve. Point A also defines the left ventricular end-diastolic pressure (LVEDP) on the y axis and the end-diastolic volume on the x axis. The distance from point A to point B demonstrates an increase in pressure at a constant volume, the period of isovolumic contraction, and the beginning of systole. Point B signifies the beginning of systolic ejection when the left ventricular pressure exceeds that of the aorta, opening the aortic valve. Ejection occurs from point B to point C and ends once the pressure in the aorta exceeds that in the ventricle, closing the aortic valve. Point C denotes both the end-systolic pressure and the beginning of diastole. The distance from point C to point D demonstrates a decrease in pressure at a constant volume, the period of isovolumic relaxation. At point D, the pressure in the atrium exceeds the ventricle and the mitral valve opens, beginning diastolic filling. Again, it is important to note that increases in diastolic volume result in a comparatively small increase in diastolic pressure in a normal heart. The width of the pressure-volume loop is a representation of stroke volume. Arterial elastance (*Ea*) is a measure of afterload and is

simply thought of as the ratio of end-systolic pressure to stroke volume. With acute inferior vena cava occlusion, the effects of sequentially smaller ventricular volumes on ventricular pressure can be diagrammed. This allows for the definition of two relationships: the end-systolic pressure-volume relationship (ESPVR), which is a measure of intrinsic contractility, and the end-diastolic pressure-volume relationship (EDPVR), which is a measure of diastolic stiffness and relaxation (Fig. 11.5). The ESPVR shifts with changes in myocardial contractility, while the EDPVR shifts as the passive relaxation properties of the ventricle change and is also dependent on the left ventricular filling pressures. We can gain additional information from the pressure-volume loop regarding ventricular work (Fig. 11.6). The area inside the pressure-volume loop estimates the stroke work of the ventricle, in practical terms the amount of energy spent for the given stroke volume. The entire area bound by the ESPVR and EDPVR is an estimate of not only the energy spent for ejection but also the potential energy stored and is an estimate of ventricular wall tension and myocardial oxygen demand.

As with the Starling and force-tension curves, the effects of changing various hemodynamic parameters can be diagrammed using a single pressure-volume loop. When afterload (denoted by the end-systolic pressure) and contractility (denoted by the slope of the ESPVR) are unchanged, a decrease in preload (demonstrated by a lower end-diastolic volume) decreases the stroke volume, while an increase in preload (demonstrated by an increase in end-diastolic volume) increases the stroke volume (Fig. 11.7A and B). This is anticipated as previously demonstrated with the Starling curve. The pressure-volume loop gives us additional important information, however. In the setting of increased preload, there is not only an increase in stroke volume but also an increase in stroke work (demonstrated by an increase in the area of the pressure-volume loop) and an increase in myocardial wall tension and oxygen consumption (demonstrated by an increase in the area of the pressure-volume relationship). Thus, the increase in stroke volume is at the expense of increased myocardial work and oxygen consumption. Similarly, we can evaluate isolated changes in afterload while maintaining similar preload (left ventricular end-diastolic volume [LVEDV] remains constant) and contractility (the ESPVR slope remains constant) (Fig. 11.8A and B). A decrease in afterload (demonstrated by a lower end-systolic pressure) results in an increased stroke volume with mild decreases in myocardial work and oxygen consumption. An increase in afterload (demonstrated by a higher end-systolic pressure) results, conversely, in a decreased stroke volume and an

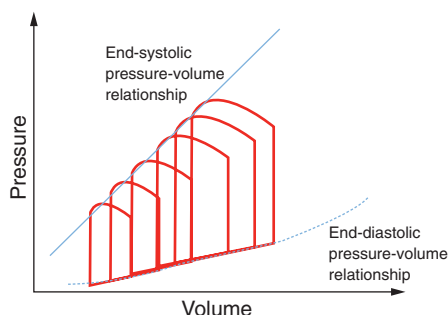


Fig. 11.5 End-systolic and end-diastolic pressure-volume relationships. Acute inferior vena cava occlusion with sequential recordings of pressure-volume loops at continually declining preloads allows for determination of the end-systolic and end-diastolic pressure-volume relationships.

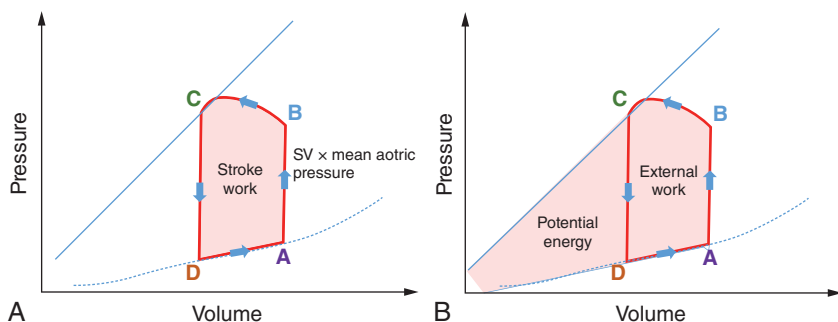


Fig. 11.6 (A) Ventricular stroke work. The area inside the pressure-volume loop is representative of ventricular stroke work. (B) Ventricular wall tension. The area of the pressure-volume relationship is a sum of the actual stroke work of the ventricle and the potential energy stored by the ventricle. This is representative of ventricular wall tension and myocardial oxygen consumption. SV, Stroke volume.

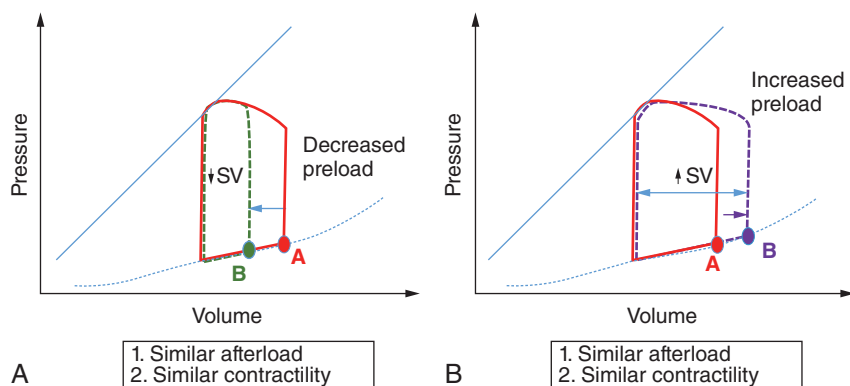


Fig. 11.7 Effects of preload changes on the pressure-volume loop with similar afterload and contractility. (A) A decrease in preload (end-diastolic volume decreases from A to B, demonstrated by the blue arrow) results in decreased stroke volume (SV), denoted by a decrease in the width of the pressure-volume loop (green loop). (B) An increase in preload (end-diastolic volume increases from A to B, demonstrated by the purple arrow) results in increased SV (blue arrow), denoted by an increase in the width of the pressure-volume loop (purple loop).

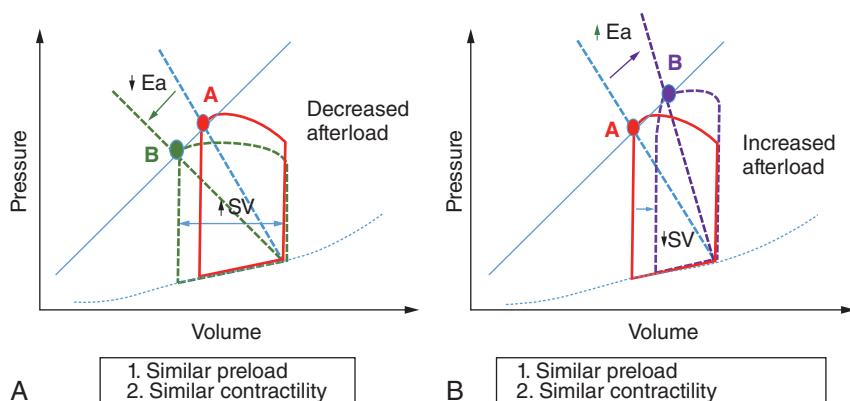


Fig. 11.8 Effects of afterload changes on the pressure-volume loop with similar preload and contractility. (A) A decrease in afterload (A to B) with similar preload and contractility results in an increase in stroke volume (SV), denoted by the width of the pressure-volume loop (blue arrow, green loop). Note the downward shift in the slope of the arterial elastance (Ea) (green arrow). (B) An increase in afterload (A to B) with similar preload and contractility results in a decrease in SV, denoted by a decrease in the width of the pressure-volume loop (blue arrow, purple loop). Note the upward shift of the slope of the Ea (purple arrow).

increased myocardial wall tension and oxygen consumption. Finally, changes in contractility can be demonstrated by means of the pressure-volume loop (Fig. 11.9). Changes in contractility are represented on the pressure-volume loop by a change in the slope of the ESPVR. Maintaining similar preload (end-diastolic volume remains constant) and afterload (end-systolic pressure remains constant), an increase in contractility results in an increase in stroke volume, myocardial stroke work, and oxygen consumption. Conversely, a decrease in contractility results in a decrease in stroke volume, myocardial work, and oxygen consumption.

With a basic background of ventricular hemodynamics, it is much easier to understand the pathophysiology of shock and heart failure and lends to a better comprehension of the effects of various therapies on the failing ventricle.

PATHOPHYSIOLOGY OF CARDIOGENIC SHOCK

Cardiogenic shock is defined as persistent hypotension, severe reduction in cardiac index, elevated filling pressures, and reduced end organ perfusion. The inadequacy of end organ perfusion, as evidenced by such things as oliguria, mental status changes, peripheral cyanosis, and cool extremities, defines the clinical presence or absence of shock

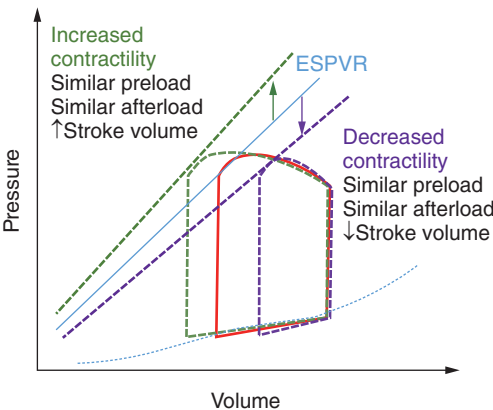


Fig. 11.9 Effects of contractility changes on the pressure-volume loop. Increases in contractility (*green loop*) result in increases in stroke volume from a baseline state (*red loop*), denoted by the increased width of the pressure-volume loop and an upward shift of the slope of the end-systolic pressure-volume relationship (ESPVR). Decreases in contractility (*purple loop*) result in decreases in stroke volume, denoted by the decrease in the width of the pressure-volume loop and a downward shift of the slope of the ESPVR.

Table 11.1 Diagnostic Criteria for Shock		
CRITERIA	MANIFESTATION	PORTENDS POOR PROGNOSIS
Hypotension	Systolic BP <90 mm Hg for 30 min, unable to reverse with fluids	Mean arterial pressure <60 mm Hg
Cardiac index	<2.2 L/min per BSA	<2.0 L/min per BSA
Peripheral vasoconstriction	Cool extremities	
Renal ischemia	Decreased urine output	<25 mL/h
Cerebral ischemia	Mental status changes	Obtunded
Normal or high filling pressures	PCWP >18 mm Hg	

BP, Blood pressure; BSA, body surface area; PCWP, pulmonary capillary wedge pressure.

(Table 11.1). Mortality secondary to cardiogenic shock is high and ranges from 40% to 80% depending on the underlying etiology. Cardiogenic shock can develop from inadequacy of output from the left ventricle, right ventricle, or both.^{3,4} A wide variety of conditions can lead to cardiogenic shock, with acute myocardial infarction and acute decompensated chronic systolic heart failure representing the most common causes (Box 11.1). Both systolic and diastolic mechanisms contribute to shock. An absolute or relative decrease in the left ventricular output leads to hypotension and decreased systemic and coronary perfusion. Increases in LVEDV and LVEDP lead to pulmonary congestion and increased wall stress. This combination results in myocardial oxygen demand that outstrips supply. The hypotension resulting from inadequate output also leads to a compensatory systemic release of catecholamines, with the goal of increasing contractility and peripheral resistance. Unfortunately, the body's efforts to maintain perfusion to vital organs often result in increased cardiac oxygen demand and work, as well as worsening of organ perfusion secondary to vasoconstriction (Fig. 11.10). Importantly, cardiogenic shock can often be reversed in the early stages, thus recognition and initiation of proper therapy in a timely manner are crucial.^{5,6}

HEMODYNAMICS OF CARDIOGENIC SHOCK

The differential diagnosis of shock is broad, and the first task of the physician is to determine whether or not the underlying cause of shock is cardiogenic. Clinical suspicion is often confirmed with the aid of right heart and left heart catheterization. The central aortic waveform has several characteristics during shock that can be helpful in its early recognition. The height of the pulse pressure tracing is directly associated with the left ventricular contractility, while the width offers

Box 11.1 Causes of Cardiogenic Shock**Acute Myocardial Infarction**

1. Pump failure
2. Right ventricular infarction
3. Ventricular septal rupture
4. Acute mitral regurgitation
5. Free wall rupture
6. Acquired left ventricular outflow tract obstruction

Arrhythmia

1. Bradycardia
2. Tachycardia

Acute Valvular Regurgitation

Aortic regurgitation
Mitral regurgitation
Acute prosthetic valve regurgitation

Obstruction

Decompensated aortic stenosis
Hypertrophic obstructive cardiomyopathy
Acquired left ventricular outflow tract obstruction
Atrial myxoma
Acute prosthetic valve obstruction

Pump Failure

Idiopathic
Myocarditis
Ischemic

Compressive

Tamponade

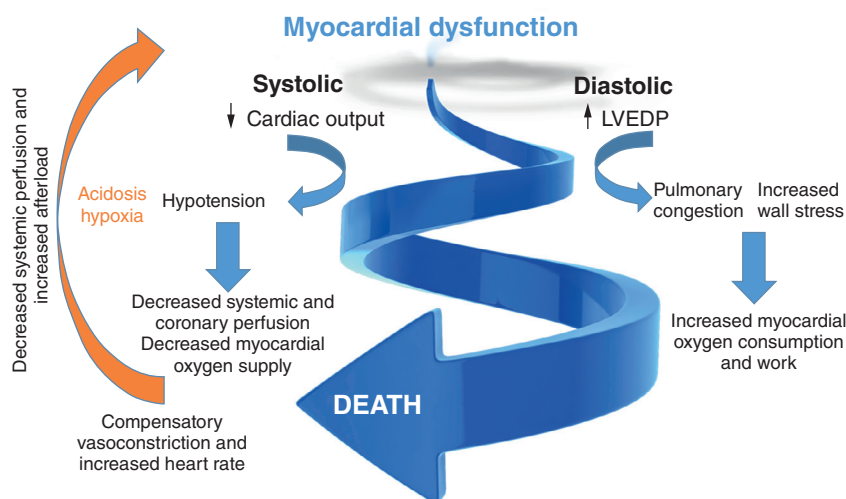


Fig. 11.10 The downward spiral of cardiogenic shock. Both diastolic and systolic abnormalities lead to decreased myocardial oxygen supply and increased myocardial oxygen demand. Compensatory mechanisms aimed at increasing systemic perfusion worsen the scenario. *LVEDP*, Left ventricular end-diastolic pressure.

information about stroke volume (Fig. 11.11). The majority of patients with cardiogenic shock display a decreased pulse pressure height and width as both contractility and stroke volume are reduced (Fig. 11.12). However, in cases where the left ventricular function is normal but inadequate, as with acute mitral regurgitation or postmyocardial infarction ventricular septal defect, the pulse pressure height may be normal, but the width reduced, suggesting a marked decrease in forward

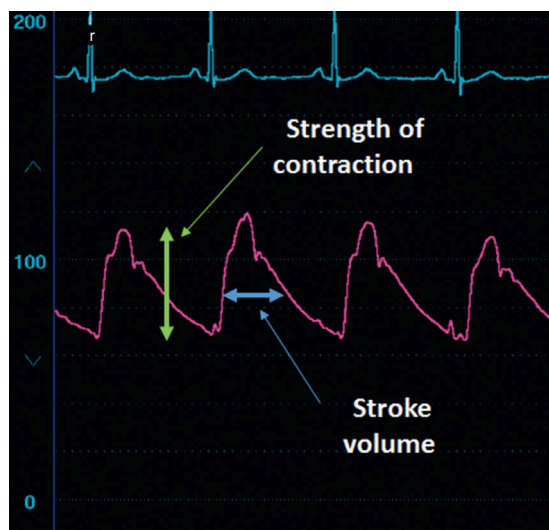


Fig. 11.11 Normal aortic waveform. Note the robust upstroke of the normal aortic waveform. The height of the waveform (green arrow), or pulse pressure, is representative of the strength of contraction. The width of the waveform (blue arrow) is representative of the stroke volume.

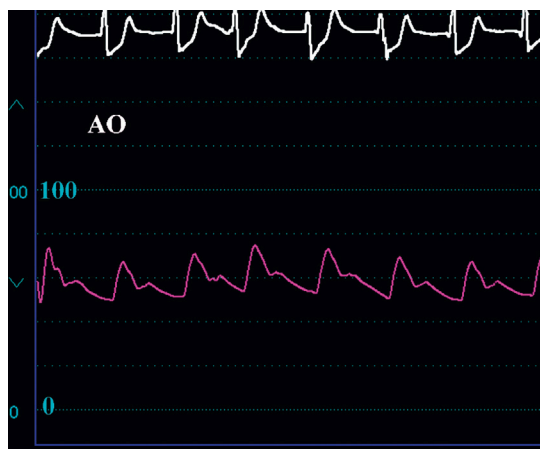


Fig. 11.12 Aortic (AO) waveform in cardiogenic shock. This is an AO pressure tracing obtained from a patient with cardiogenic shock demonstrating a low systolic pressure, a reduced pulse pressure, and narrowed waveform.

stroke volume (Fig. 11.13A). This narrow “spiked” appearance can herald cardiovascular collapse even in the patient with a “normal” pulse pressure height and apparently adequate systolic pressure (see Fig. 11.13B).

Right heart catheterization can be used to distinguish cardiogenic shock from other types of shock. Elevation in the right-sided filling pressures, in particular the pulmonary capillary wedge pressure (PCWP), is a consistent feature of cardiogenic shock, but other clues to etiology may be present. For example, acute severe mitral regurgitation often demonstrates a large, pathologic *v* wave in the PCWP tracing. It is important to remember that some causes of cardiogenic shock may have a low PCWP. For example, shock caused by right-sided heart failure will demonstrate low cardiac output, hypotension, and a normal to low PCWP. Cardiac tamponade in the setting of hypovolemia may also demonstrate low to normal filling pressures, although equalization of diastolic pressures will be present. In combination with aortic pressure measurement, the systemic vascular resistance (SVR) can be calculated. The SVR is high in the presence of cardiogenic shock due to compensatory peripheral vasoconstriction, with the exception of the end stages of shock when acidosis may lead to profound vasodilation.

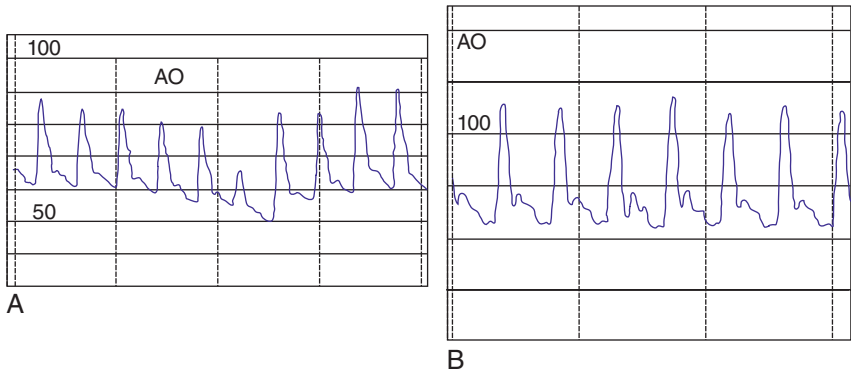


Fig. 11.13 (A) Aortic (AO) waveform in acute mitral regurgitation. In cases where the left ventricular function is normal but inadequate, as with acute mitral regurgitation, the pulse pressure height may be normal but the width reduced, suggesting a marked decrease in forward stroke volume. (B) AO waveform with an acute ventricular septal defect. In the setting of an acute ventricular septal rupture postmyocardial infarction, the peak systolic pressure is normal. However, note the narrow pulse pressure width and a *spiked appearance* of the AO pressure tracing. Note also the location of the aortic valve closure notch.

The SVR is easily calculated using the following formula:

$$\text{SVR} = \frac{(\text{Mean aortic pressure}) - (\text{Mean right-atrial pressure})}{\text{Cardiac output}}$$

Multiplying the SVR by 80 converts to resistance units, dynes per s/cm^5 . This can be indexed to body size by multiplying by body surface area:

$$\text{SVRI} = (\text{SVR}) \times (\text{Body surface area}),$$

where SVRI is the SVR indexed to body size.

The normal SVR = 1170 ± 270 dynes per s/cm^5 , and the normal SVRI = 2130 ± 450 dynes per $\text{s}/\text{cm}^5 \times \text{m}^2$.

Myocardial work and oxygen demand are significantly increased in the presence of shock. Numerous factors have been shown to contribute to increased myocardial oxygen consumption, including many aspects discussed above: contractility, preload, and afterload. While myocardial demand is increased during shock, supply is often reduced. In the face of elevated left ventricular diastolic pressures, coronary blood flow, particularly in the microvasculature, is compromised. Increases in the left ventricular mass from various underlying conditions can further worsen the flow in the microvasculature in the setting of shock. Additionally, in the face of hypotension, systemic diastolic pressures are lower, decreasing coronary filling. Increases in heart rate shorten diastolic time, further decreasing coronary blood flow and myocardial oxygen supply.

Recent studies have also shown that cardiac power output (CPO) can provide prognostic information in the setting of heart failure and cardiogenic shock. CPO is calculated by the following formula:

$$\text{CPO} = \frac{\text{Cardiac output} \times \text{Mean arterial pressure}}{451}$$

This measure takes into account not only cardiac output but also the ability of the heart to generate systemic flow and blood pressure. In other words, cardiac output alone cannot generate end organ perfusion. Maintenance of adequate arterial pressure is also necessary. For example, in end-stage heart failure, while cardiac output seems adequate to maintain perfusion, there is severe vasodilation present secondary to metabolic disarray that cannot be overcome by cardiac output alone. This measure has been shown to predict mortality in the setting of shock due to myocardial infarction, nonischemic cardiomyopathy, and acute myocarditis. Additionally, CPO is predictive of the patients who will progress to worsening heart failure before the development of shock. A CPO cutoff of less than 0.6 Watts (W) has been shown to have the best sensitivity and specificity for predicting worsening heart failure in patients admitted for heart failure, and a cutoff of less than 0.53W predicts mortality in patients admitted with cardiogenic shock.⁷⁻¹⁰ Similarly, the pulmonary artery pulsatility index (PAPi) can be used to predict the right ventricular failure after inferior myocardial infarction or left ventricular support device implantation. PAPi is easily calculated by:

$$\text{PAPi} = (\text{PASP} - \text{PADP}) / \text{RAP}$$

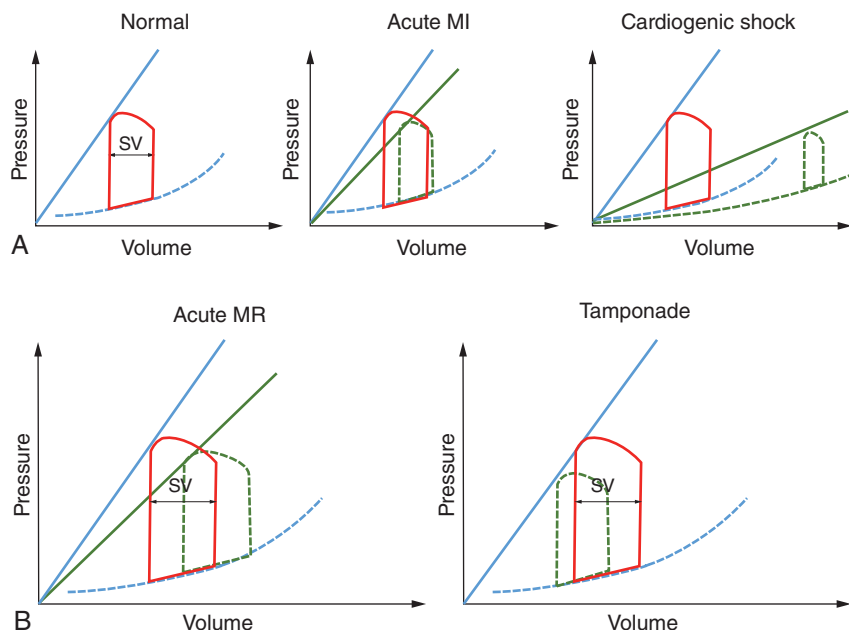


Fig. 11.14 (A) Pressure-volume loops in a normal heart and various causes of cardiogenic shock. Acute myocardial infarction (MI) results in a decrease in contractility (decreased slope of the end-systolic pressure-volume relationship), a decreased stroke volume (SV) (decreased width of the pressure-volume loop), and an increase in end-diastolic volume. Progression to cardiogenic shock leads to further decreases in contractility and increases in end-diastolic volumes. (B) Pressure-volume loops in acute mitral regurgitation (MR) and tamponade. Acute MR results in marked increases in end-diastolic volumes and a simultaneous fall in forward cardiac output even though SV may remain similar. The abrupt increase in ventricular filling may also acutely reduce contractility. Cardiac tamponade results in decreased cardiac output as a result of marked decreases in end-diastolic volumes. This results in a decreased SV and cardiac output and a fall in the mean arterial pressures, despite the maintenance of ventricular contractility. Red loops represent a normal heart. Green loops represent the pressure-volume loop associated with each pathophysiology. (A, From Rijal CS, Naidu SS, Givertz MM, et al. 2015 SCAI/ACC/HFSA/STS clinical expert consensus statement on the use of percutaneous mechanical circulatory support devices in cardiovascular care. *J Am Coll Cardiol*. 2015;65(19):e7–e26.)

where PASP is the pulmonary artery systolic pressure, PADP is the pulmonary artery diastolic pressure, and RAP is the right atrial pressure.¹¹ Data from the National Cardiogenic Shock Initiative suggest that a PAPI of <0.9 and CPO of <0.6 suggest the need for right and left ventricular support, respectively, in the setting of acute myocardial infarction.¹²

It is important to understand the effects of various causes of shock on the pressure-volume loop to understand what hemodynamic alterations will be most helpful for the patient. In the setting of a large, acute myocardial infarction, there is an abrupt decrease in contractility, as demonstrated by a decrease in the slope of the ESPVR. This results in an acute decrease in stroke volume (Fig. 11.14A). If the mechanism of shock remains unchecked, the hemodynamics may further degenerate, with the eventual development of cardiogenic shock. At this point, in an attempt to maintain the stroke volume, there is a marked increase in end-diastolic volume, or preload. There is now also a marked decrease in contractility worsened by the compensatory increase in the SVR.

The effects of other shock etiologies on the pressure-volume loop may look very different from acute myocardial infarction, necessitating different therapies (see Fig. 11.14B). In the presence of acute severe mitral regurgitation, there is a marked increase in end-diastolic volumes because of increased blood return from the left atrium. There is a simultaneous fall in forward cardiac output, even though stroke volume may remain similar. Due to reduced forward stroke volume, the mean arterial pressures fall. The abrupt increase in ventricular filling may also acutely reduce contractility, although contractility may be increased in the face of increased ventricular filling. Conversely, in cardiac tamponade, there is a marked decrease in end-diastolic volumes, resulting in a decreased stroke volume and fall in the mean arterial pressures despite the maintenance of ventricular contractility.

PATHOPHYSIOLOGY OF HEART FAILURE WITH REDUCED EJECTION FRACTION

Approximately 5.7 million Americans carry the diagnosis of heart failure, with about 550,000 new cases per year. More than 11 million physician visits per year occur for a heart failure diagnosis, and more than half of individuals with a

Box 11.2 Causes of Systolic Heart Failure With Reduced Ejection Fraction**Coronary Artery Disease**

- Hypertension
- Infection
- Viral, parasitic, fungal, bacterial

Infiltrative Disease

- Amyloidosis, hemochromatosis

Toxins

- Chemotherapeutic agents, alcohol, cocaine, amphetamines, environmental toxins

Endocrine and Metabolism Disorders

- Hypothyroidism, pheochromocytoma, diabetes, nutrient deficiencies

Rheumatologic Disorders

- Giant cell arteritis, systemic lupus erythematosus, scleroderma, sarcoidosis
- Electrolyte abnormalities
- Tachycardia induced
- Peripartum
- Familial, genetic

diagnosis of heart failure will die within 5 years. The most common causes for chronic systolic heart failure are coronary artery disease and hypertension. However, there are numerous other causes of systolic heart failure. Many cases remain idiopathic with no identified etiology (Box 11.2). Heart failure with reduced ejection fraction can result from a chronic progression of an underlying process such as hypertension or may result from an acute process such as following a viral infection.¹³

HEMODYNAMICS OF HEART FAILURE WITH REDUCED EJECTION FRACTION

The hemodynamic findings of chronic heart failure can range from almost entirely normal in compensated heart failure to significantly abnormal in the face of decompensation. As the ventricle dilates, increasing preload could generate greater myocardial stretch and thus contraction and stroke volume. However, this is at the expense of greater energy demand because of increased myocardial work. In decompensated heart failure, LVEDP elevates; it is also reflected in an elevation of the PCWP. This results in a passive elevation in the pulmonary artery (PA) and right-sided heart pressures. As pressure and volume increase and the ventricle stretches, the relatively rigid pericardium plays a greater role in determining this pressure-volume relationship. In chronic heart failure with the left ventricular enlargement, the constraining effect of the pericardium couples the ventricles via the intraventricular septum. In other words, changes in volume in one chamber influence the opposing chamber. This phenomenon is used to explain the mechanism by which diuresis improves the left ventricular stroke volume in chronic heart failure. The Frank-Starling mechanism states that a decrease in chamber volume (preload) would *decrease* the stroke volume; however, diuresis *improves* the stroke volume when there is cardiac chamber enlargement and high filling pressures because diuresis decreases the right ventricular chamber volume; this allows an increase in the left ventricular diastolic volume, thereby improving the left ventricular stroke volume.¹⁴ With acute heart failure, the abrupt increase in cardiac chamber volume within the confines of a relatively inelastic pericardium may cause a hemodynamic finding similar to constrictive pericarditis.¹⁵

Severe left ventricular dysfunction is associated with several abnormalities on the left ventricular pressure waveform. In patients with normal systolic function, left ventricular upstroke, or the rate of left ventricular pressure rise, is brisk with a rapid decline after reaching peak systolic pressure. With the onset of diastole, the left ventricular pressure is normally very low (zero or even negative); it rises throughout diastole to reach an end-diastolic pressure of 10–12 mm Hg. Patients with severe left ventricular dysfunction may show a slow rise in the left ventricular pressure because of poor contractility that yields a triangular appearance to the left ventricular waveform (Fig. 11.15). In addition, the left ventricular diastolic pressure is high early in diastole, with LVEDP reaching very high levels (40–50 mm Hg). A prominent *a* wave is often seen on the left ventricular tracing because of atrial contraction against an already high end-diastolic volume within a noncompliant ventricle (Fig. 11.16). In patients with heart failure the LVEDP is often used as a surrogate for the mean left atrial pressure; however, a prominent *a* wave may interfere with the accurate estimation of the left atrial pressure. In such cases the left ventricular pre-*a* wave pressure, or the mean left ventricular diastolic pressure, correlates well with the mean left atrial pressure.¹⁶

A large *v* wave may be seen on the PCWP tracing in severe heart failure without associated mitral regurgitation. Normally, at low pressures and volumes, the small additional increase in volume that occurs during passive left atrial filling when the mitral valve is closed causes just a small rise in the pressure and a physiologic *v* wave. In heart failure,

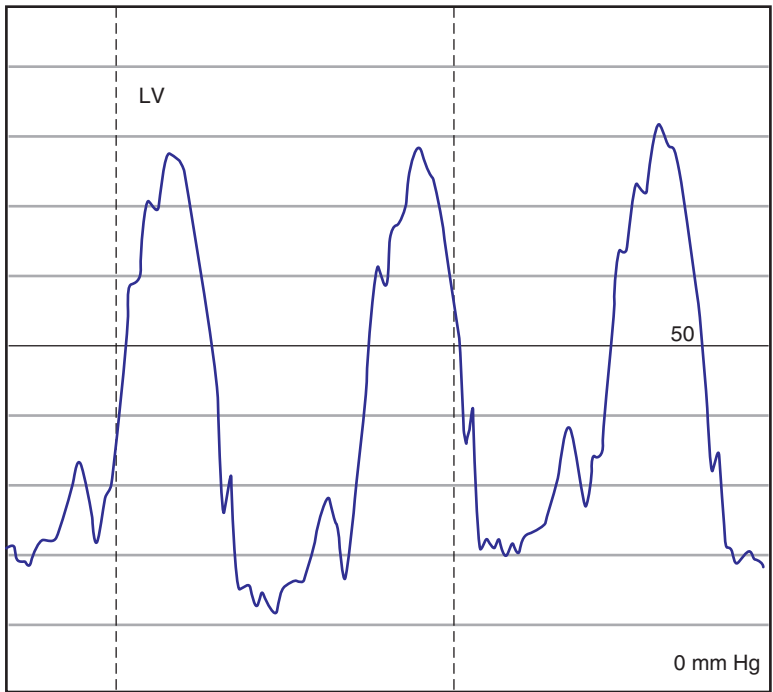


Fig. 11.15 Left ventricular waveform in severe left ventricular dysfunction. Note the delayed upstroke and pressure decay leading to a triangular appearance of the waveform. *LV*, Left ventricle.

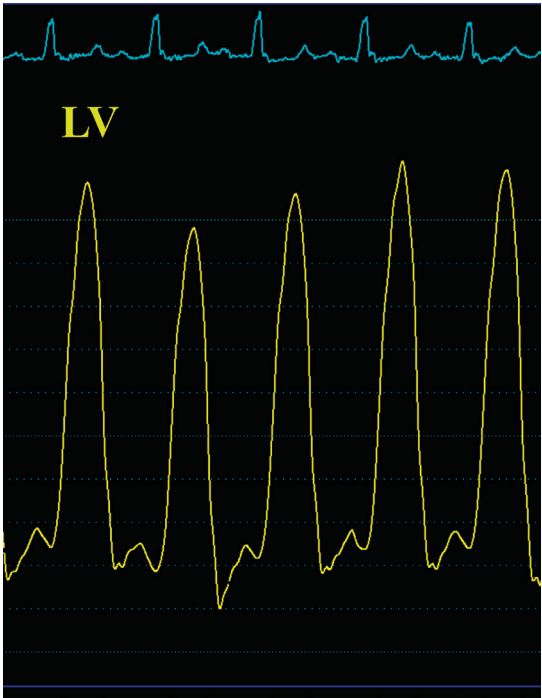


Fig. 11.16 Left ventricular waveform in heart failure, prominent *a* wave. Example of a large *a* wave on a left ventricular pressure waveform in a patient with heart failure. *LV*, Left ventricle.



Fig. 11.17 Left ventricular waveform in heart failure, prominent *v* wave. Prominent *v* wave on a pulmonary capillary wedge pressure tracing in a patient with heart failure from severe left ventricular systolic dysfunction and no mitral regurgitation.



Fig. 11.18 Aortic waveform in heart failure, pulsus alternans. Pulsus alternans on an aortic pressure tracing in a patient with severe left ventricular dysfunction.

the distended volumes of the left atrium shift the compliance curve to the right so that the additional volume from passive atrial filling causes a large increase in the pressure, generating a substantial *v* wave (Fig. 11.17).

Pulsus alternans represents an unusual and interesting hemodynamic sign of a failing heart. The sign is manifest by an alternating rise and fall in aortic systolic pressure from beat to beat while the patient is in a regular cardiac rhythm (Fig. 11.18). This case is different from a *pulsus bigeminus*, which is seen with an atrial or ventricular bigeminal rhythm, or a *pulsus paradoxus*, in which the rise and fall of aortic systolic pressure is related to the respiratory cycle. Pulsus alternans usually implies serious underlying myocardial disease. It occurs in association with severe aortic stenosis, cardiomyopathy, and severe coronary artery disease. However, it has been described in normal hearts after the termination of a supraventricular tachycardia. It may be accentuated by decreasing venous return, such as occurs with inferior vena caval occlusion, and may become more apparent after a premature ventricular beat. The pattern of alternating rise and fall in systolic pressure may be seen on the left ventricular, right ventricular, and PA waveforms, in addition to the aortic waveform (Fig. 11.19). Both right ventricular and pulmonary arterial pulsus alternans may be seen independent of left-sided pulsus alternans if there is isolated right heart failure. Isolated atrial alternans has also been described. The cause of pulsus alternans is not entirely clear. One explanation suggests that it might be due to alternation of myocardial contractility on a beat-to-beat basis because of a loss of the number of cells that contract on alternate beats because of abnormal intracellular calcium cycling—that is, a form of *localized* electrical mechanical dissociation.¹⁷

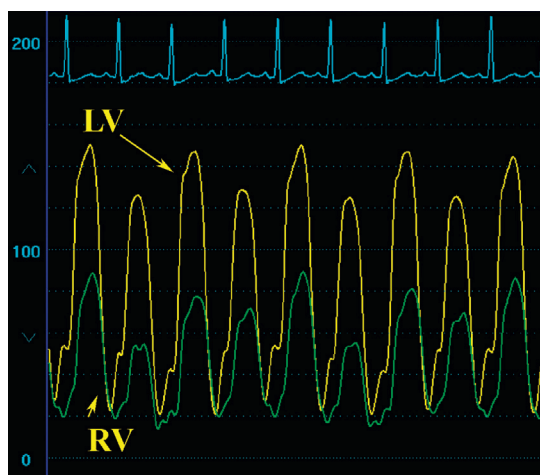


Fig. 11.19 Left ventricular and right ventricular waveforms in heart failure. These tracings were obtained from the same patient as Fig. 11.18 and demonstrate pulsus alternans of the left ventricle (LV) and right ventricle (RV).

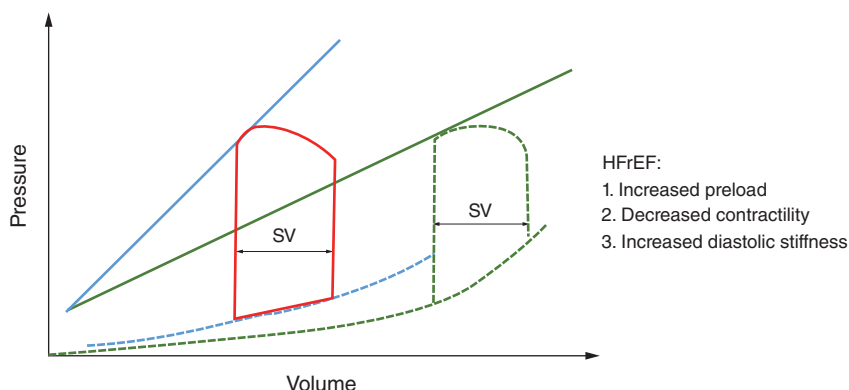


Fig. 11.20 Pressure-volume loop in heart failure with reduced ejection fraction (HFrEF). Preload is increased (increased end-diastolic volume) with the goal of maintaining stroke volume (SV) in the face of reduced contractility (decreased slope of end-systolic pressure-volume relationship). There is an increase in diastolic stiffness, demonstrated by an upward shift in the end-diastolic pressure-volume relationship. The green loop represents the failing heart. The red loop represents a normal heart.

Pressure-volume loops can also clarify the hemodynamics of acute decompensated systolic heart failure, or heart failure with reduced ejection fraction. When the heart fails in a more chronic manner, again preload is increased, with the goal of maintaining stroke volume in the face of reduced contractility. This is demonstrated by a shift of the pressure-volume loop to the right and a decrease in the slope of the ESPVR. There is also a marked increase in diastolic stiffness, demonstrated by an upward shift in the EDPVR (Fig. 11.20).

HEMODYNAMICS OF THERAPIES FOR HEART FAILURE AND SHOCK

Based on the understanding of hemodynamics during cardiogenic shock and decompensated heart failure, it is evident that the goals of supportive therapy would be to increase both cardiac output and systemic perfusion. This would be ideally done by creating circumstances whereby energy supply to the heart is maximized and energy demand is minimized. This section will review many of the current therapies available for cardiogenic shock and decompensated heart failure and their respective hemodynamic effects.

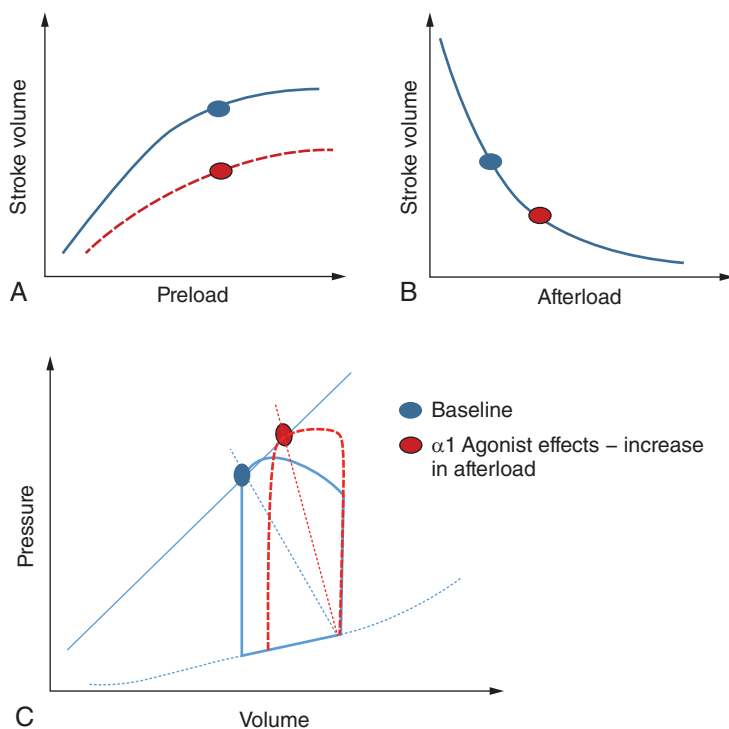


Fig. 11.21 Effects of phenylephrine on the Starling curve, force-tension curve, and pressure-volume loop. (A) The Starling curve demonstrates a constant preload with reduced stroke volume because of either a decrease in contractility or an increase in afterload (shift from the blue oval to the red oval). (B) The force-tension curve demonstrates a clear increase in afterload (shift from the blue oval to the red oval). (C) The blue loop, oval, and dotted line represent the baseline pressure-volume loop, left ventricular end-systolic pressure, and arterial elastance (Ea), respectively. The pressure-volume loop demonstrates an increase in afterload (increased slope of Ea, red dotted line) with a resultant decrease in stroke volume (decreased width of pressure-volume loop, red loop) with other parameters remaining constant.

INOTROPES AND VASOACTIVE DRUGS

While pressure-volume loops may not be routinely used in clinical medicine, it is critical to understand both the advantageous and disadvantageous hemodynamic effects of the various treatment modalities available for cardiogenic shock. Treatment of cardiogenic shock aims to maintain adequate tissue perfusion and reverse the downward spiral of hemodynamic derangement, while addressing the underlying cause or inciting event. Many drugs used in the treatment of cardiogenic shock have been shown to increase morbidity and mortality.^{18–20} With the use of pressure-volume loops, it is easy to demonstrate with the following examples why this may be the case. Phenylephrine (Fig. 11.21, red oval) is often used to increase afterload and systemic pressure and is a pure afterload-increasing agent with only α_1 agonist effects. This is demonstrated on the Starling curve as a downward shift of the curve, indicating either a decrease in preload or an increase in afterload. The force-tension curve confirms an increase in afterload. The pressure-volume loop gives additional information, demonstrating a decrease in the stroke volume and an increase in the myocardial oxygen consumption and wall tension, as evidenced by the decreased width of the pressure-volume loop and an increase in the area of the pressure-volume relationship, respectively. Dobutamine, a drug often used to increase contractility and cardiac output, has both β_1 and β_2 effects (Fig. 11.22). The β_2 effects result in skeletal muscle vasodilation, decreasing afterload demonstrated by an upward shift of the Starling curve, and a decrease in afterload on the force-tension curve (green oval). Similarly, the pressure-volume loop demonstrates a decrease in afterload (decreased slope of Ea) and an increase in stroke volume (green loop). Note, in this case, the myocardial oxygen demand is decreased (decreased area of the pressure-volume relationship). The addition of β_1 effects increases contractility, as evidenced by a further upward shift of the Starling curve at a consistent preload and upward shift of the force-tension curve (red oval). The pressure-volume loop demonstrates increased contractility (increased slope of the ESPVR), increased stroke volume (width of the pressure-volume loop), increased myocardial stroke work (increased area of the pressure-volume loop), and an overall increase in myocardial oxygen consumption (increased area of the pressure-volume relationship) (red loop). Thus, improvement in hemodynamics with the use of dobutamine comes with

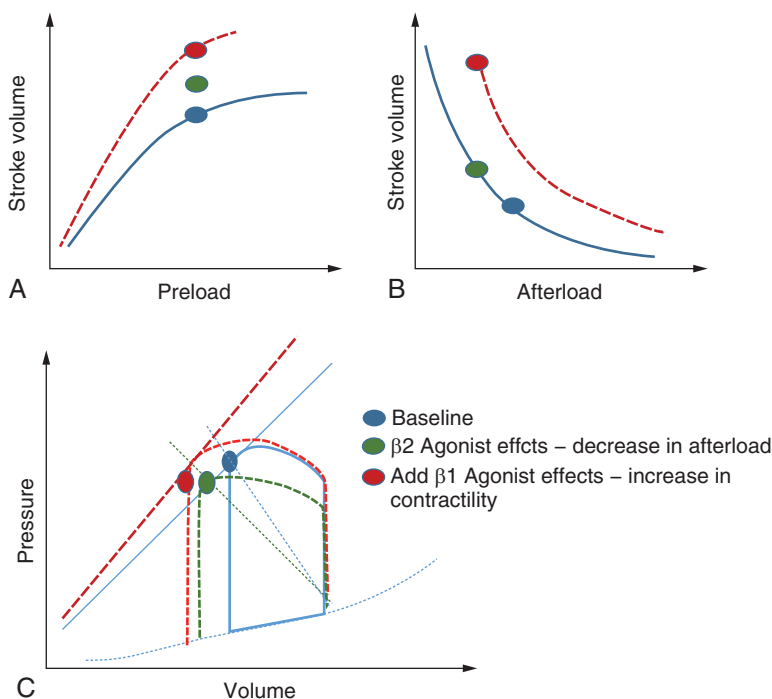


Fig. 11.22 Effects of dobutamine on the Starling curve, force-tension curve, and pressure-volume loop. (A) The Starling curve demonstrates a constant preload with increases in stroke volume because of decreases in afterload (*green oval*) due to β_2 effects. Increases in contractility result in further upward shift of the Starling curve (*red oval*) due to β_1 effects. (B) The force-tension curve demonstrates a decrease in afterload (*green oval*). Increases in contractility shift the curve upward (*red oval*). (C) The pressure-volume loop demonstrates a decrease in afterload (*green loop*). Addition of increased contractility increases the slope of the end-systolic pressure-volume relationship (*red loop*). This results in increased stroke volume (increased width of the pressure-volume loop), increased myocardial work (increased area of the pressure-volume loop), and increased myocardial oxygen consumption (increased area of the pressure-volume relationship).

the cost of potentially marked increases in myocardial oxygen demand and work. The β_1 effects can further contribute to arrhythmia generation, further increasing the oxygen demand and, because of tachycardia, reducing the oxygen supply. Finally, norepinephrine is classically used in the treatment of cardiogenic shock and has both α_1 effects to increase the blood pressure and β_1 effects to increase contractility, both potentially desired in the setting of shock (Fig. 11.23). The effects of increased afterload are demonstrated by a downward shift of the Starling curve and an increase in afterload on the force-tension curve (*green oval*). This is demonstrated on the pressure-volume loop with an increase in the end-systolic pressure and a narrowing of the width of the pressure-volume loop and thus a decrease in stroke volume (*green loop*). Increases in contractility then shift the Starling curve and the force-tension curve upward (*red oval*). The pressure-volume loop demonstrates an increase in stroke volume (increased width of the pressure-volume loop) and contractility (increased slope of the ESPVR) at the expense of increased myocardial stroke work (increased area of the pressure-volume loop) and increased oxygen consumption (increased area of the pressure-volume relationship) (*red loop*). In addition to the effects on left ventricular hemodynamics, many vasoactive drugs may have deleterious effects on end organ perfusion depending on the amount of systemic vasoconstriction provided by the drug.

MECHANICAL LEFT VENTRICULAR SUPPORT DEVICES

There are now several percutaneous devices available for left ventricular support in the setting of cardiogenic shock, including the intraaortic balloon pump (IABP), the Impella percutaneous left ventricular assist device (LVAD) (Abiomed), the TandemHeart LVAD (LivaNova), and extracorporeal membrane oxygenation (ECMO). The mechanism by which each functions and the hemodynamic effects, complications, and contraindications will be discussed in the following section.^{6,21–23}

Intraaortic Balloon Counterpulsation

The IABP is often the first device to be used when inotropic and vasoactive support are insufficient because of its ease of use and insertion.^{3,22–25} The balloon catheter is inserted percutaneously, generally through an 8-French (Fr) sheath

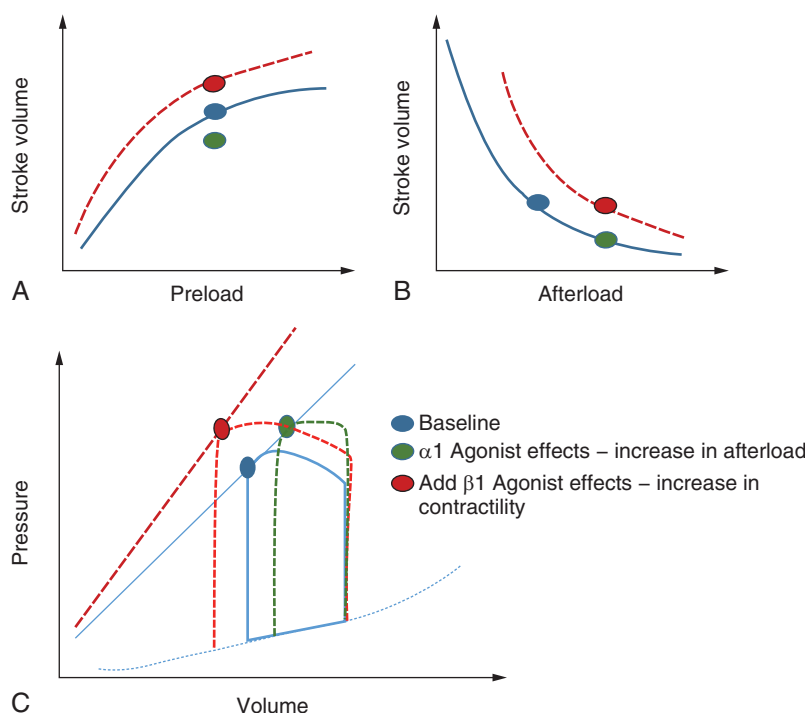


Fig. 11.23 Effects of norepinephrine on the Starling curve, force-tension curve, and pressure-volume Loop. (A) The Starling curve demonstrates a downward shift (green oval) due to an increase in afterload and a much larger upward shift due to increase in contractility (red oval). (B) The force-tension curve demonstrates an increase in afterload (green oval). Addition of increased contractility shifts the curve upward (red oval). (C) The pressure-volume loop demonstrates an increase in afterload (green oval and loop). Addition of increased contractility increases the slope of the end-systolic pressure-volume relationship (red oval). This results in increased stroke volume (increased width of the pressure-volume loop), increased myocardial work (increased area of the pressure-volume loop), and increased myocardial oxygen consumption (increased area of the pressure-volume relationship) (red loop).

via the common femoral artery into the descending aorta (Fig. 11.24). The balloon is timed to inflate during diastole against a closed aortic valve, displacing aortic blood (30–50 mL, depending on the balloon volume) and increasing the diastolic pressure within the aorta (*diastolic augmentation*). The balloon rapidly deflates at the end of diastole, just prior to aortic ejection, creating a potential space of 30 to 50 mL within the aorta, lowering the impedance of the aorta, reducing end-diastolic pressure, and resulting in afterload reduction. Thus the IABP provides two major beneficial effects in cardiogenic shock: improved tissue perfusion by diastolic augmentation of aortic pressure and enhancement of ventricular function by reducing afterload. It is important to recognize that systolic pressures may decrease further because of afterload reduction. The net effect, however, is an increase in the mean arterial pressure, thereby improving tissue perfusion. Cardiac output and stroke volume mildly increase while heart rates are either unaffected or may decrease slightly as a consequence of improved hemodynamics. The pressure-volume loop clearly demonstrates these effects with a decrease in afterload (E_a), an increase in stroke volume (width of the pressure-volume loop), and no change in contractility (the slope of the ESPVR remains constant). Diastolic ventricular parameters also improve (decreased preload) with a downward shift of the EDPVR curve (Fig. 11.25). Cardiac output generally increases on average about 0.5 L/min.

One of the presumed hemodynamic benefits of the IABP relates to the potential enhancement of coronary blood flow, a mostly diastolic phenomenon. However, the effect on coronary blood flow varies depending on the underlying hemodynamics and the degree of coronary arterial narrowing. Using the Doppler flow wire in human subjects, studies suggest that aortic counterpulsation increases coronary blood flow velocity in the proximal, prestenotic segment of the coronary artery and in the distal artery, only if no significant stenosis is present. However, in the presence of a severe coronary stenosis, coronary blood flow velocity in the distal artery does not increase with the IABP. This suggests that the dramatic relief of ischemia afforded by the balloon pump in patients with critical coronary stenosis is from afterload and preload reduction and improvement in diastolic hemodynamics, thereby decreasing the myocardial oxygen demand rather than via a direct increase in coronary blood flow.^{26–30}

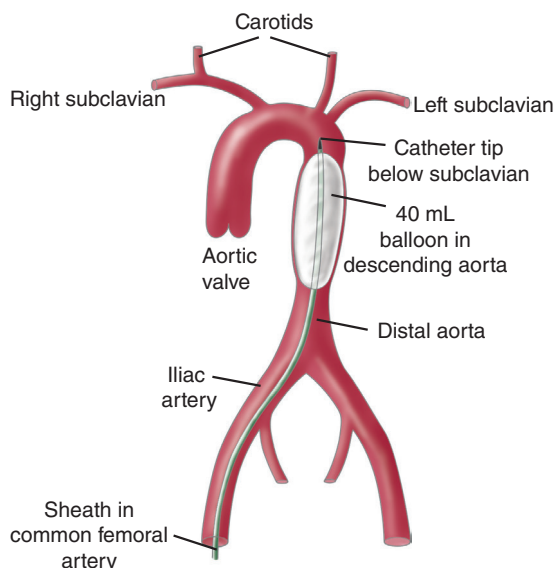


Fig. 11.24 Correct intraaortic balloon pump placement. This schematic represents proper placement of an intraaortic balloon pump in the descending aorta via the right femoral artery. The tip of the catheter should be below the subclavian artery, usually achieved by placing it at the tracheal bifurcation.

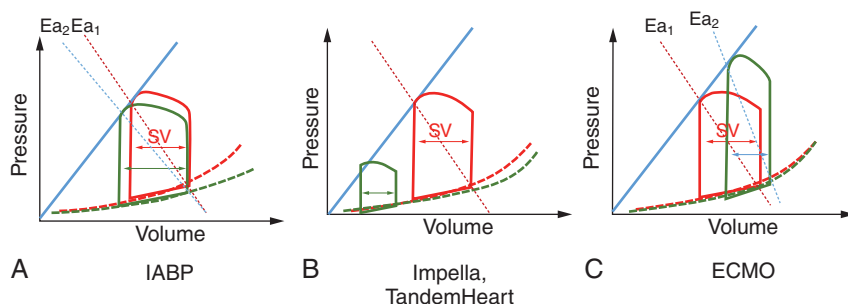


Fig. 11.25 Pressure-volume loops of ventricular support devices. The red loop represents the baseline condition. The green loop represents the effect of the support device. (A) The intraaortic balloon counterpulsation provides a decrease in afterload and a resulting small increase in stroke volume (SV). (B) The Impella provides significant decreases in myocardial work and myocardial oxygen demand. The TandemHeart has the ability to do this under optimal conditions (see the text for full explanation). (C) Extracorporeal membrane oxygenation (ECMO) results in marked increases in afterload and a subsequent decrease in SV. There is also a significant increase in myocardial oxygen consumption (increased area of the pressure-volume relationship). *Ea*, Arterial elastance; *IABP*, intraaortic balloon pump.

The hemodynamic benefits vary in any individual and depend on the degree of ventricular impairment, the underlying cardiac rhythm, the SVR, and the timing and volume of balloon inflation. The balloon pump should be thought of as a minimal support device; it augments cardiac output but is less effective in patients with profoundly depressed cardiac output. Similarly, patients with excessive tachycardia have diminished diastolic filling times, decreasing the efficacy of the device. Patients with cardiogenic shock and low SVR will have little or no additional afterload reduction and are unlikely to achieve any diastolic pressure augmentation. For this reason, the IABP is not an effective treatment for septic shock or other, noncardiogenic causes associated with low SVR. It is also important to note that the right ventricular function sufficient for adequate cardiac output is essential. Finally, the optimal benefits are seen only when the timing of inflation and deflation is correct and when the appropriate volume of inflation is used; improper timing or inadequate inflation volumes may not provide beneficial effects and may, in the event of improper timing, be detrimental.

Timing of inflation and deflation can best be accomplished by placing the pump on a 1:2 mode (inflated once for every two cycles) (Fig. 11.26). Onset of inflation optimally occurs just after the diastolic notch. Deflation should be

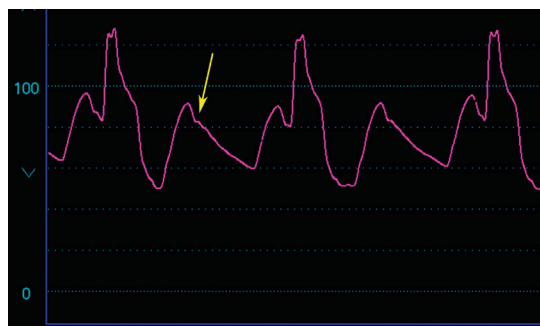


Fig. 11.26 Proper intraaortic balloon pump timing. The pump is set at a 1:2 ratio, meaning that it is set to inflate and deflate every other cardiac cycle. The balloon inflates just after the diastolic notch on the aortic pressure waveform (*arrow*). Deflation should be completed before aortic ejection begins on the next cardiac cycle. There is a lower end-diastolic pressure of the augmented beat compared with the unaugmented beat.

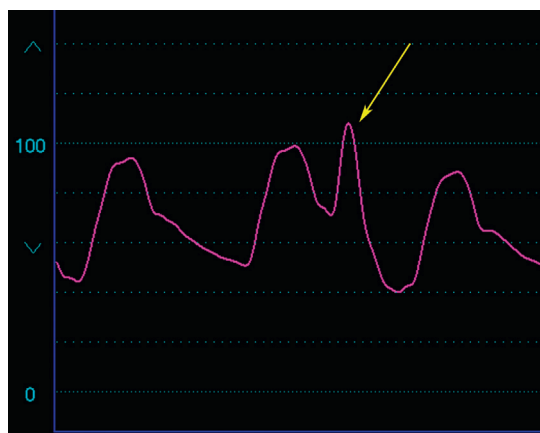


Fig. 11.27 Improper intraaortic balloon counterpulsation timing. This example demonstrates early deflation as well as late inflation. The balloon timing is set at a 1:2 ratio. Note the weak appearance of the augmented beat (*arrow*) providing little additional diastolic pressure.

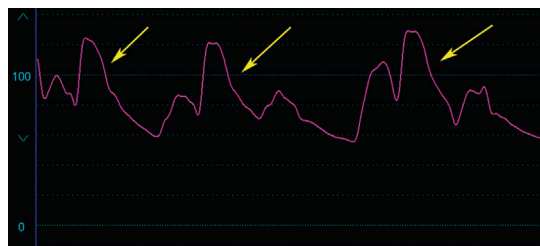


Fig. 11.28 Improper intraaortic balloon counterpulsation timing. This example demonstrates late deflation. The balloon timing is set at a 1:2 ratio. The arrows point to the augmented beats. Note that the end-diastolic pressure of the augmented beat is much higher than that of the unaugmented beat. This indicates that the balloon remains partially inflated at the beginning of aortic ejection, increasing afterload, with potentially detrimental effects.

completed prior to the beginning of aortic ejection, as evidenced by a lower end-diastolic pressure for the augmented beat compared with the unaugmented beat. Recognizing faulty timing is important because improper timing may, at minimum, fail to achieve a therapeutic effect, or worse, cause clinical deterioration (Figs. 11.27–11.29). Early deflation or late inflation of the balloon causes inadequate augmentation and either no or poor additional perfusion pressure. Late

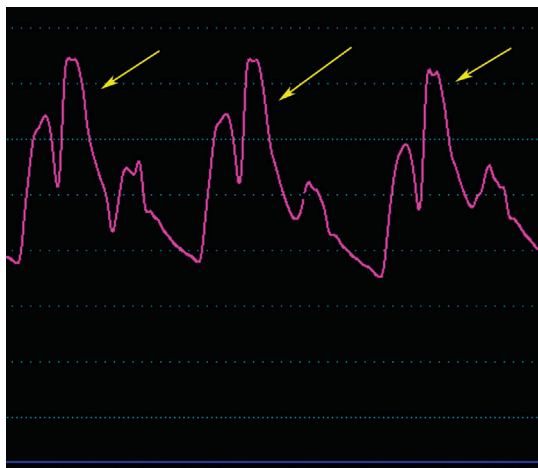


Fig. 11.29 Improper intraaortic balloon counterpulsation timing. This is an example of early inflation and late deflation. The balloon pump timing is set at a 1:2 ratio. The arrows point to the augmented beat. Balloon inflation occurs before the dicrotic notch. Also, the end-diastolic pressure of the augmented beat is higher than that of the unaugmented beat, consistent with late deflation.

Box 11.3 Intraaortic Balloon Pump Synopsis

Intraaortic Balloon Pump

- **Insertion:** Inserted via an 8-French femoral sheath
- **How it works:** Inflates during diastole, deflates during systole
- Increases aortic diastolic and mean pressure—augments diastolic flow
- Decreases aortic systolic pressure—reduces afterload
- Slight increase in cardiac output/stroke volume (0.5 L/min)
- Useful if full support not needed
- **Contraindications:** Aortic insufficiency and severe peripheral vascular disease
- **Relative contraindications:** Aortic aneurysm, severe iliac tortuosity
- **Complications:** Bleeding, peripheral ischemia, thromboembolism, compartment syndrome, aortic dissection, vascular injury, balloon rupture, or entrapment

deflation or early inflation may be harmful. In the case of late deflation, increased impedance and increased afterload are present because of ejection against a partially inflated balloon. In the event of early inflation, the aortic valve has not yet closed; the inflating balloon may cause aortic regurgitation and increased impedance with adverse consequences.

The IABP is indicated as initial support for most causes of cardiogenic shock and can be effective at stabilizing hemodynamics in patients with mechanical complications of acute myocardial infarction (acute mitral regurgitation and ventricular septal defect).^{31,32} It may not benefit patients with shock from right ventricular infarction, unless associated with a large left ventricular infarction and elevated PCWP, causing increased right ventricular afterload. The balloon pump improves the hemodynamic and rhythm status of patients with shock from malignant ventricular arrhythmia.³³ In cardiogenic shock from ischemic pump failure, the balloon pump improves clinical stability but appears most useful when coupled with revascularization.^{34,35} Patients with decompensated aortic stenosis or shock from acute mitral regurgitation also improve, and in patients with end-stage heart failure, the balloon pump provides an initial step in a bridge to transplantation or a ventricular assist device. Contraindications and complications are reviewed in [Box 11.3](#).

Impella Percutaneous LVAD

The Impella device represents the only percutaneously placed device to utilize an intracorporeal pump for the left ventricular support. Impella offers various levels of support depending on the device used: Impella CP or Impella 5.5, which offer cardiac output augmentation of around 4.0 and 6.0 L/min, respectively. Each device has 10 performance levels (P levels), 0 to 8 and “boost,” which dictate the revolutions per minute of the device and thus the cardiac output that can be achieved. The CP device is inserted percutaneously into the femoral artery using a 14-Fr sheath. The larger 5.5 device requires a 23-Fr sheath typically requiring vascular surgical cut down, either femoral or axillary. The catheter

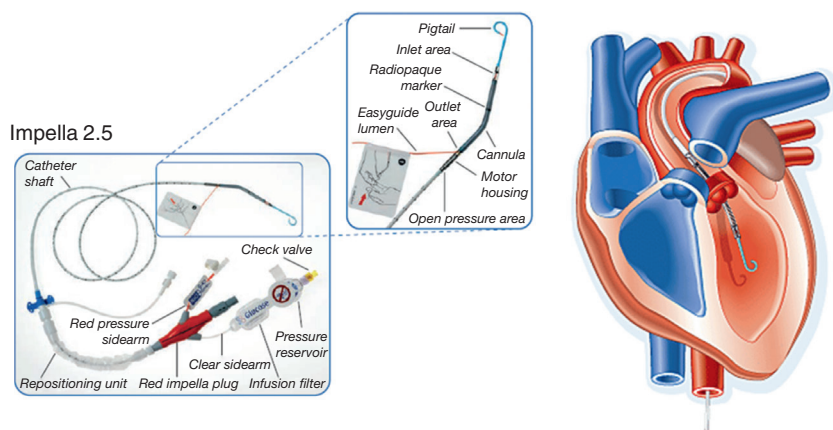


Fig. 11.30 The Impella left ventricular support device. The catheter houses a microaxial pump that sits within the ventricle (inlet area) and delivers blood in a continuous manner to an opening in the ascending aorta just above the aortic valve (outlet area). This is known as left ventricle-to-aorta bypass. (Courtesy Abiomed, Danvers, Massachusetts.)

itself is reminiscent of a left ventricular pigtail catheter and is inserted into the ventricle over a guidewire in a similar manner. The catheter houses a microaxial pump that sits within the ventricle (inlet area) and delivers blood continuously to an opening in the ascending aorta just above the aortic valve (outlet area) and is known as a left ventricle-to-aorta bypass (Fig. 11.30). The Impella console displays a placement signal that allows the operator to confirm that the device is properly placed within the ventricle. A purge fluid of dextrose solution is continuously infused through the device to prevent blood from entering the motor while the catheter is in place. The Impella console guides the operator through all steps of the setup process. Systemic anticoagulation following device sheath placement is essential.

The Impella offers several hemodynamic modifications that are beneficial in cardiogenic shock and advanced heart failure.^{3,7,21–23,36} Unlike the IABP, it does not depend on the intrinsic rhythm of the heart for timing and function. However, it does depend on the right ventricular function sufficient for adequate cardiac output. Because of its ability to directly remove blood from the left ventricle, the left ventricular wall stress can be markedly reduced with a simultaneous increase in systemic flow and mean arterial pressure. This allows for increased coronary blood flow during diastole because of the continuous nature of the pump. Its largest effect on coronary blood flow, however, is due to an improvement in coronary microvascular flow in the setting of a decreased LVEDP and LVEDV.³⁷ Thus there is an increase in the myocardial oxygen supply with a decrease in the myocardial oxygen demand, an ideal outcome for a patient in cardiogenic shock. Indeed, of the devices discussed, it is the most effective in achieving left ventricular unloading, and its antegrade method of blood return does not contribute to ventricular “reloading” as occurs with the TandemHeart and ECMO devices discussed later. The left ventricle-to-aorta support strategy used by Impella is the first to demonstrate a reduction in myocardial oxygen demand below baseline.^{7,38} Impella provides a significantly greater increase in cardiac index compared with the IABP (0.49 vs. 0.11, $P = .02$) and allows for a greater decrease in vasoactive drugs compared with the IABP. CPO also increases to a greater degree because of increases in both cardiac output and mean arterial pressure, as demonstrated in the ISAR-SHOCK (Efficacy of Study of LV Assist Device to Treat Patients with Cardiogenic Shock) trial using the 2.5 device in the setting of myocardial infarction and cardiogenic shock.³⁹ Impella 2.5 has also been shown to provide superior hemodynamic support during high-risk percutaneous intervention compared with the IABP, although there was no apparent beneficial effect on 30-day adverse events.⁴⁰ As with the IABP, it may not benefit patients with shock from right ventricular infarction, unless associated with a large left ventricular infarction and elevation in PCWP causing increased right ventricular afterload. The beneficial hemodynamic effects of the Impella are elegantly demonstrated using the pressure-volume loop (Figs. 11.25 and 11.31). The entire pressure-volume loop is shifted significantly to the left. Not only is there a marked decrease in LVEDP and LVEDV, but there is also a downward shift of the left ventricular EDPVR curve, indicating improvement in ventricular relaxation properties. The area within the pressure-volume loop and the area bound by the pressure-volume relationship are significantly decreased, indicating a decrease in cardiac work and myocardial oxygen demand, respectively. Increasing levels of support show further decreases in ventricular demand (see Fig. 11.31). Note that the Impella can provide support only to a maximum of 6.0 L/min. Stroke volume provided by the ventricle is decreased as the device is providing much of the ventricular output. This is demonstrated in the arterial waveform in a case of high-risk percutaneous coronary intervention in which the only coronary artery supplying the entire myocardium is being manipulated. Prior to balloon inflation, the aortic waveform is pulsatile with a mean arterial pressure of 66 mm Hg. During balloon inflation (causing coronary obstruction and ischemia and thus further diminishing ventricular contractile function), the aortic waveform is flattened, but the mean arterial pressure remains near 70 mm Hg, with the majority of cardiac output now being supplied by the Impella device (Fig. 11.32).

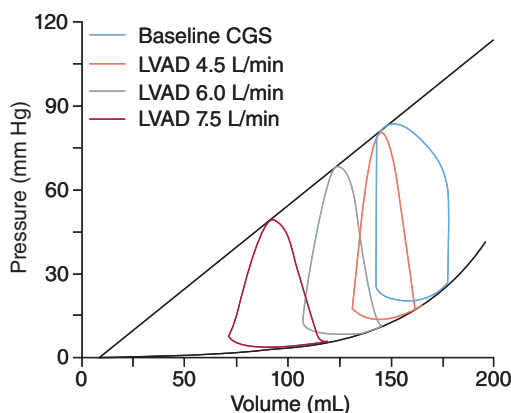


Fig. 11.31 Pressure-volume loops generated by the Impella device and left ventricle to aorta support. Pressure-volume loops demonstrate decreased ventricular work and myocardial oxygen demand with increasing levels of support. Note that the Impella can only provide up to 6.0 L/min of support. Colored loops correspond to LVAD flows in the legend. CGS, Cardiogenic shock; LVAD, left ventricular assist device. (From Burkhoff D, Sayer G, Doshi D, Uriel N. Hemodynamics of mechanical circulatory support. *J Am Coll Cardiol*. 2015;66(23):2663–2674.)

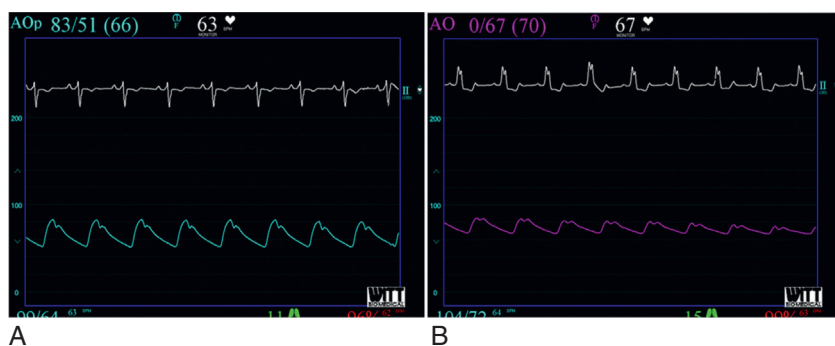


Fig. 11.32 Arterial waveforms during high-risk percutaneous coronary intervention (PCI) with the use of Impella. (A) Baseline arterial waveform with Impella support. The initial arterial waveforms in a case of high-risk PCI in which the only artery supplying the myocardium is being manipulated. Note a low systolic pressure with a relatively normal pulse pressure and mean arterial pressure of 66 mm Hg. (B) Arterial waveform with Impella support during balloon inflation. After coronary balloon inflation, the aortic waveform is flattened, but the mean arterial pressure remains near 70 mm Hg, with the majority of cardiac output now being supplied by the Impella device.

The operator must be familiar with troubleshooting several aspects of device operation. Correct placement of the pump is essential for proper function. The placement and motor current signals displayed by the console allows the operator to determine if the outlet area of the catheter is in the aorta as appropriate, or completely within the ventricle. Correct placement will demonstrate both an aortic (red) and ventricular (white) waveform on the placement screen with a pulsatile motor current (green) (Fig. 11.33). If the catheter is completely within the aorta, an aortic placement signal (red) will be displayed and the motor current (green) will appear flat. (Fig. 11.34). Similarly, if the catheter is completely in the ventricle, a ventricular placement signal (red) will be displayed with a flat motor current (green) (Fig. 11.35). The motor current is a measure of energy intake of the device and varies with motor speed and the pressure difference between the inlet and outlet valves. When the Impella is correctly positioned, the motor current will be pulsatile, signifying the pressure differences between the inlet and outlet valves, and will vary at points in the cardiac cycle. The placement signal and motor current can be used together to assure that the device maintains its proper position. Should the catheter position migrate, the console will alarm to indicate inappropriate catheter placement. If the operator chooses to use the placement screen on the console, it should be noted that if the majority of cardiac output is being provided by the device, the arterial waveform will appear flattened, and the placement signal will not be accurate in determining

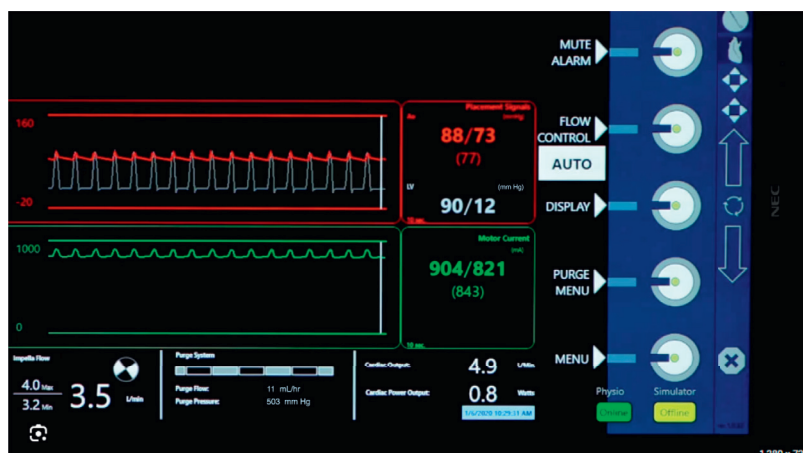


Fig. 11.33 Impella console denoting the correct position of the catheter. The placement signal shows the location of the catheter. The aortic signal is shown in red, and the ventricular signal in white. The motor current signal (*green*) is pulsatile when the catheter is in the correct location. The console also displays the minimum, maximum, and average Impella flows, the cardiac output, and the cardiac power output. (Courtesy Abiomed, Danvers, Massachusetts.)

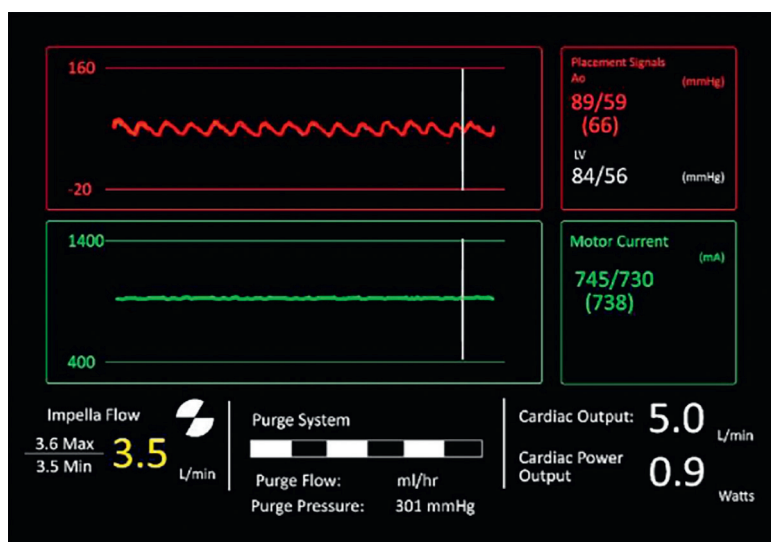


Fig. 11.34 Impella console showing Impella position in the aorta. Note the motor current is flat. (From Gottula AL, et al. Impella in transport: physiology, mechanics, complications, and transport considerations. *Air Med J.* 2021;41(1):114–127, Fig. 8b.)

the device position. In addition to the catheter being properly placed in the ventricle, the operator must assure that the outlet valve does not sit too close to the aortic valve as this will increase the likelihood of hemolysis, aortic regurgitation, and valvular damage. Finally, the maintenance of purge flow and pressure are critical to the proper functioning of the device. The controller automatically regulates the pressure of the purge fluid delivered to the catheter. An alarm will sound when the pressure falls below 300 mm Hg or exceeds 1100 mm Hg. A low pressure generally indicates a leak in the system, a problem with the purge cassette, or damage to the catheter itself. A high purge pressure may indicate a kink in the tubing or need for a lower concentration of dextrose solution. As with any LVAD, decreases in flow may be due to relative hypovolemia resulting in catheter suction events, too deep of a device implantation in the left ventricle resulting in catheter suction, bleeding, tamponade, right ventricular failure, or thrombosis.

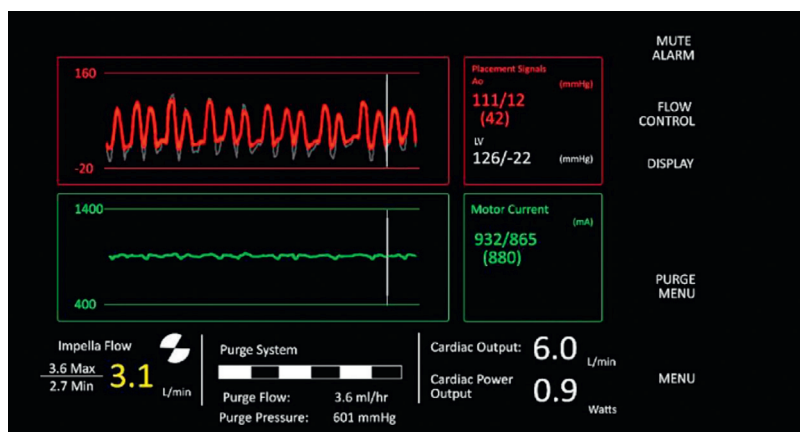


Fig. 11.35 Image of Impella console showing Impella position in the ventricle. Note the motor current is no longer pulsatile. (From Gottula AL, et al. Impella in transport: physiology, mechanics, complications, and transport considerations. *Air Med J.* 2021;41(1):114–127.)

Box 11.4 Impella Left Ventricular Support Device Synopsis

Impella Left Ventricular Support Device

- **Insertion:** Retrograde across aortic valve, approved 10 days (CP) and 30 days (5.5)
- CP: 4.0 L/min, femoral or axillary, 14-French sheath
- 5.5: 6.0 L/min, axillary or femoral cut down, 23-French sheath
- **How it works:** Pulls blood from the left ventricle and expels the blood into the ascending aorta, continuous flow
- Direct ventricular unloading—decreases end-diastolic volume and pressure
- Decreases wall tension
- Decreases mechanical work
- Increases diastolic (coronary) blood flow
- Increases cardiac output and mean arterial pressure
- Requires patient to have sufficient oxygenation
- **Contraindications:** Aortic stenosis or insufficiency, severe peripheral vascular disease, mechanical aortic valve, ventricular septal defect, left ventricular thrombus, inability to anticoagulate
- **Complications:** Bleeding, peripheral ischemia, aortic valve injury, hemolysis, vascular injury, transient ischemic attack/stroke, ventricular arrhythmias, ventricular perforation, aortic insufficiency

The Impella device is approved for use in the United States for up to 10 days for the CP device and for up to 30 days for the 5.5 device. It is important to understand the several contraindications to use this specific device. Mural thrombus in the left ventricle is a strong contraindication to this device, and a transthoracic echocardiogram should be obtained prior to its placement. The device should also not be used in patients with a mechanical aortic valve and should be used with great caution in patients with aortic stenosis, especially if valve area is less than 0.6 cm². The presence of a ventricular septal defect may also increase right-to-left shunting as the Impella decreases the left ventricular volume and pressure. Systemic anticoagulation must be used with Impella; therefore the device should not be used in patients who cannot tolerate a high level of systemic anticoagulation. As with the IABP, moderate-to-severe aortic insufficiency may be worsened. Complications and contraindications are listed in [Box 11.4](#).

TandemHeart Percutaneous LVAD

The TandemHeart is one of several available LVADs that utilize an extracorporeal pump. The device essentially bypasses the left ventricle by removing blood directly from the left atrium and delivering the blood in a retrograde manner to the

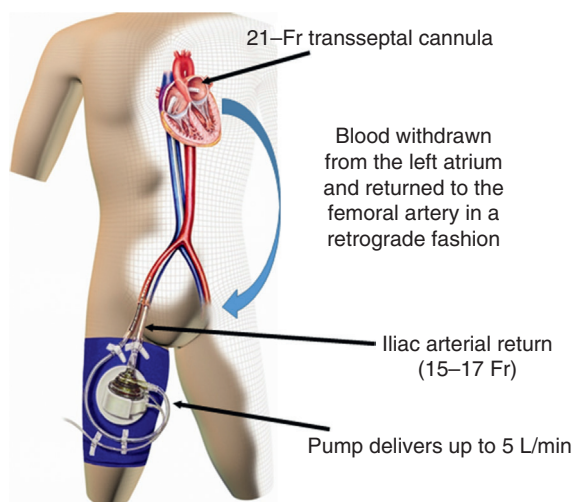


Fig. 11.36 The TandemHeart left ventricular support device. The device removes oxygenated blood directly from the left atrium and delivers the blood in a retrograde manner generally to the femoral artery via an external continuous-flow centrifugal pump. This is known as *left atrial-to-aorta bypass*. The pump is designed to deliver up to 5 L/min at a maximum speed of 7500 rpm. Fr, French. (Courtesy Liva Nova, London, United Kingdom.)

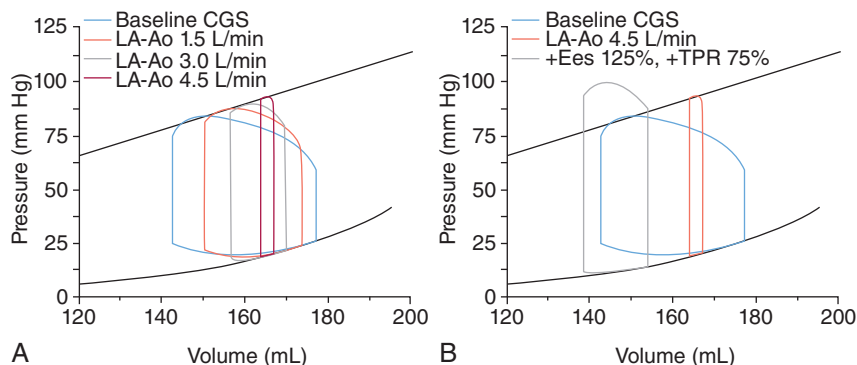


Fig. 11.37 Pressure-volume loops generated by the TandemHeart device. The blue loop represents baseline cardiogenic shock (CGS). (A) Pressure-volume loops demonstrating increasing levels of ventricular support (colored loops correspond with left ventricular assist device [LVAD] flows in the legend). (B) The red loop represents maximum LVAD support. The addition of afterload reduction (total peripheral resistance [TPR]) and contractility increases (end-systolic elastance [Ees]) improve the hemodynamics of the TandemHeart pump (red loop shifts to gray loop). LA-Ao, Left atrial-to-aorta bypass. (From Burkhoff D, Sayer G, Doshi D, Uriel N. Hemodynamics of mechanical circulatory support. *J Am Coll Cardiol*. 2015;66(23):2663–2674.)

femoral artery via an external continuous-flow centrifugal pump (left atrial-to-femoral artery bypass). For long-term placement the subclavian or axillary arteries can be used. The inflow cannula is a 21-Fr sheath inserted in the femoral vein and placed via a transseptal approach into the left atrium. The outflow cannula is placed into the femoral artery and is available in 15- and 17-Fr sizes. The size chosen generally depends on the patient's arterial size, with the larger cannula providing potential for greater increases in cardiac output. The pump is designed to deliver up to 5 L/min at a maximum speed of 7500 rpm (Fig. 11.36). Proper implantation and maintenance of the pump is generally facilitated by the assistance of a perfusionist. Systemic anticoagulation is required with a recommended activated clotting time of 400 at device insertion and greater than or equal to 200 throughout the support period.^{7,21,22}

The TandemHeart offers several hemodynamic modifications that are beneficial in cardiogenic shock and advanced heart failure (Figs. 11.25 and 11.37). Like the Impella, it also does not depend on the intrinsic rhythm of

the heart for timing and function but does depend on the right ventricular function sufficient for adequate cardiac output. Because of its ability to indirectly remove blood from the left ventricle via the left atrium, the LVEDP and thus left ventricular wall stress can be reduced, similar to the Impella. However, the retrograde delivery of blood results in increased mean arterial pressure (afterload) which must be overcome by increased left ventricular pressure generation. The ventricle often achieves this via increases in left ventricular end-systolic volumes. Thus while myocardial work decreases, the myocardial oxygen demand and the area under the pressure-volume relationship may not change significantly (see Fig. 11.37A). Additional efforts to decrease total peripheral resistance or increase end-systolic elastance (contractility) move the hemodynamic effects of the TandemHeart much closer to those of the Impella, although myocardial work remains higher (see Fig. 11.37B, *gray loop*; Fig. 11.25). Similar to Impella, because of the nonpulsatile delivery of flow at relatively high flows, the arterial tracing may appear flat with no evidence of aortic valve opening. The degree of pulsatility will depend on the inherent contractility of the ventricle as well as the amount of blood remaining in the ventricle after removal from the left atrium by the transseptal cannula. Sequential increase in flow from the device may decrease the LVEDV to the point that left ventricular stroke volume contributes little to cardiac output (see Fig. 11.37A, *red loop*). Therefore, only the mean arterial pressure of the patient should be monitored, as systolic and diastolic pressures may be altered and not reliable for hemodynamic assessment. Its largest effect on coronary blood flow, similar to Impella, is due to an improvement in microvascular flow during diastole in the setting of a decreased LVEDP and LVEDV. This results in both a decrease in the myocardial oxygen demand and an increase in the myocardial oxygen supply. The mean arterial pressure is also increased by the continuous retrograde delivery of blood through the femoral artery, further increasing the myocardial oxygen supply. TandemHeart has been shown to decrease mortality in a small case series in the setting of severe cardiogenic shock and to improve organ function prior to transplantation or permanent device therapy.^{41,42} In patients who have failed therapy with inotropes and IABP, TandemHeart has the ability to rapidly reverse terminal hemodynamic compromise and improve end organ function.⁶ As with the IABP, it may not benefit patients with shock from right ventricular infarction, unless associated with a large left ventricular infarction and elevation in PCWP that causes increased right ventricular afterload.

The operator must be familiar with troubleshooting several aspects of device operation. Initial device preparation is critical as the device must be completely devoid of air. Similar to Impella, the controller unit provides vital information (Fig. 11.38). Pump flow, as opposed to left ventricular output or total flow, is displayed by the controller and does not account for native left ventricular ejection. A PA catheter must be used to determine total flow as the mixed venous saturation will reflect both left ventricular output and pump flow. Calculations of cardiac index and SVR should be made using total flow measured by the PA catheter. Additionally, at higher device outputs, the arterial tracing may appear



Fig. 11.38 TandemHeart console. The TandemHeart console displays device flow and revolutions per minute. (Courtesy of LivaNova.)

Box 11.5 TandemHeart Left Ventricular Support Device Synopsis**TandemHeart Left Ventricular Support Device**

- **Insertion:** Requires femoral venous and arterial access, approved 24 hours to 30 days
- Arterial: 15- to 17-French femoral sheath
- Venous: 21-French transseptal sheath
- Need for intubation and potential paralytics
- **How it works:** Pulls blood from the left atrium and returns to the femoral artery in a continuous retrograde manner
- Increases cardiac output (up to 5.0 L/min depending on arterial cannula size)
- Increases the mean arterial pressure
- Indirect ventricular unloading
- Requires patient to have sufficient oxygenation
- **Contraindications:** Severe peripheral vascular disease, intraatrial devices, known atrial thrombus, Greenfield filter (relative), inability to anticoagulate
- **Complications:** Bleeding, peripheral ischemia, hemolysis, vascular injury, transient ischemic attack/stroke, cardiac tamponade, cannula dislodgement

flat, and the mean arterial pressure should be followed as opposed to systolic arterial pressure. Dislodgment of the left atrial cannula must be recognized immediately. A decrease in blood oxygen saturation and a change of blood color in the pump from bright to dark red indicate migration of the cannula into the right atrium. The outflow tubing must be immediately clamped and the pump turned off. The controller can also alert the operator should flow through the pump decrease. As with any LVAD, a decrease in flow may be caused by relative hypovolemia resulting in catheter suction events or from the device being too deeply implanted in the left atrium resulting in catheter suction; both of these are manifest by "chatter" in the catheter. Additional causes of low flow include bleeding, tamponade, right ventricular failure, or thrombosis.

The TandemHeart device is approved for use in the United States for short-term extracorporeal support. Contraindications include thrombus in the left atrial appendage, and many operators place the device with the assistance of a transesophageal echocardiogram for this reason. A transesophageal echocardiogram can also aid with the performance of the transseptal puncture. Systemic anticoagulation must be used with TandemHeart immediately after transseptal puncture; therefore the device should not be used in patients who cannot tolerate systemic anticoagulation. As with the IABP and Impella, moderate-to-severe aortic insufficiency may be worsened. Complications and contraindications are listed in Box 11.5.

Venoarterial ECMO Device

A venoarterial ECMO is a temporary extracorporeal mechanical support device, which has the ability to support not only the left ventricular function but also the lung function. Cannulation sites may vary based on the age of the patient and the time at which ECMO is initiated. In adults, venous cannulae are most commonly placed in the internal jugular or femoral vein or directly into the right atrium. Arterial cannulae are most commonly placed in the femoral or axillary artery or directly into the aorta if placed at the time of an open chest procedure. The ECMO circuit utilizes an external heat exchanger and membrane oxygenator, or "artificial lung," to provide oxygenation and ventilation. Blood drains from the venous system into the membrane oxygenator and is delivered back to the body in a retrograde manner with the use of a centrifugal pump (Fig. 11.39). The amount of oxygenation that can be provided is determined by flow rate, whereas ventilation can be adjusted by gas flow through the oxygenator. In adults, cannula sizes are similar to the TandemHeart system. The use of ECMO requires the support of an experienced perfusionist, and systemic anticoagulation is required.

ECMO offers several hemodynamic modifications that are beneficial in severe cardiogenic shock and decompensated heart failure. Like the Impella and TandemHeart, ECMO does not depend on the intrinsic rhythm of the heart for timing and function. An additional advantage of this device compared with the other support devices is that it also does not depend on adequate right ventricular function or lung function. While ECMO can provide "complete cardiorespiratory support," there are additional important and potentially detrimental effects as demonstrated by the pressure-volume loop (Figs. 11.25 and 11.40A). First, it must be noted that ECMO does not "unload" the heart; there is no significant decrease in the LVEDP as seen with Impella and TandemHeart, and this is a direct consequence of blood being removed from the venous circulation rather than the left atrium or left ventricle. This absence of ventricular unloading will have important implications, discussed later. Additionally, afterload is increased as demonstrated by an increase in the slope of E_a , and the stroke volume of the left ventricle is decreased as demonstrated by a decrease in the width of the pressure-volume loop. Notably, there is a significant increase in the myocardial oxygen demand as demonstrated by increases in the area of the pressure-volume relationship. The greater the delivery of cardiac output, the greater potential for worsening left ventricular hemodynamics (see Fig. 11.40A). ECMO has been shown to be useful in situations of prolonged cardiac arrest, even when placed during cardiopulmonary resuscitation. Because of its ease of insertion

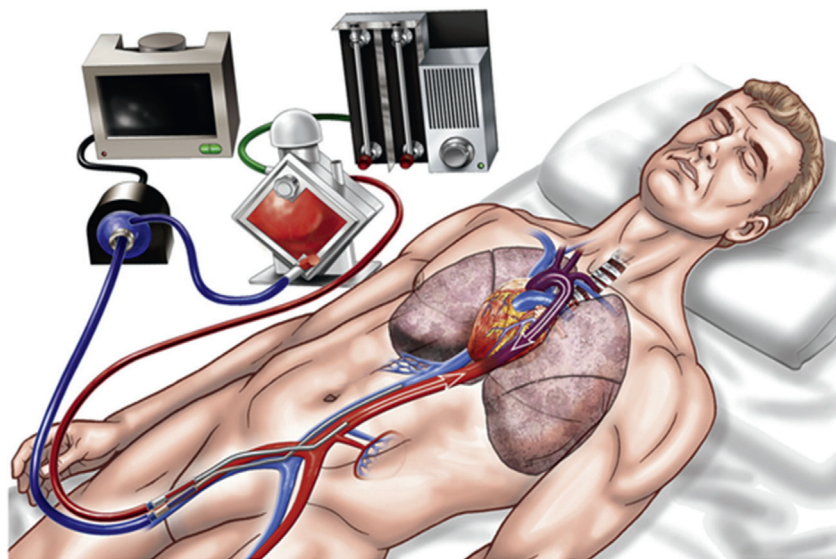


Fig. 11.39 Extracorporeal membrane oxygenator support device. The device removes blood from the right atrium and delivers oxygenated blood to the femoral artery, bypassing both the left ventricle and the lungs. Reinfused oxygenated blood (red arrow) flowing retrograde through the aorta may meet resistance from poorly oxygenated blood flowing anterograde from the left ventricle (purple arrow). Depending on the amount of cardiac function, the location of the interface between anterograde and retrograde flow will vary, and the reinfused oxygenated blood may not reach the cerebral and coronary vascular beds. (From Abrams D, Combes A, Brodie D. Extracorporeal membrane oxygenation in cardiopulmonary disease in adults. *J Am Coll Cardiol*. 2014;63(25, part A):2769–2778.)

compared with TandemHeart, full support can be achieved in a timely manner.^{43,44} Survival is also improved in postpericardiotomy patients who have an option for bridge to recovery, transplant, or permanent LVAD.^{45,46}

The operator must have a working knowledge of potential issues related specifically to ECMO. Left ventricular output must be carefully monitored during ECMO because of the increase in preload supplied by the device. A decrease in pulsatility of the arterial pressure waveform may indicate worsening left ventricular function. Inotropes or IABP support may be added to increase contractility and decrease afterload, respectively, in attempts to prevent ventricular embarrassment. Decreases in total peripheral resistance (afterload), such as with the use of an IABP, can significantly improve hemodynamics, shifting the pressure-volume loop to the left and increasing the left ventricular stroke volume. While myocardial work may not decrease and may increase, there is a significant decrease in myocardial oxygen demand as demonstrated by a decrease in the area of the pressure-volume relationship and an increase in native left ventricular stroke volume (see Fig. 11.40B). Inotropic support may also be added to increase contractility and stroke volume with resultant decreases in end-systolic and, consequently, end-diastolic volumes. This, however, is at the cost of an increase in both myocardial work and myocardial oxygen demand, as demonstrated by an increase in the area of the pressure-volume loop and pressure-volume relationship (see Fig. 11.40C). If left ventricular function still cannot be maintained, left ventricular decompression may be necessary to prevent pulmonary hemorrhage because of marked increases in the LVEDP. This can be accomplished with either an Impella catheter (ECPella) or direct left atrial or left ventricular venting. It is also important to note that left ventricular ejection may prevent oxygenated blood returned via the femoral artery from reaching the cerebral and coronary circulations (see Fig. 11.39). Thus, it is important to measure oxymoglobin saturation in the upper extremities, as that in the lower extremities may be significantly higher if the lower extremity is used for blood return. Thrombus formation within the device must be recognized immediately and can be identified by an increase in pressure gradient across the oxygenator. Thrombus formation requires immediate circuit exchange. As with other assist devices, poor pump flow may also result from relative hypovolemia, bleeding, or device thrombosis.

ECMO is classically used in patients with cardiac arrest, severe cardiogenic shock, failure to wean from cardiopulmonary bypass following cardiac surgery, as a bridge to transplant or permanent LVAD, and massive pulmonary embolism. As ECMO is often used in sicker patients who require hemodynamic as well as respiratory support (as in biventricular heart failure), survival rates are lower and decrease with the duration of device implantation. Survival in patients requiring ECMO ranges from 20% to 60% and varies depending on the underlying reason for device placement.^{47–51} Importantly, if a patient demonstrates the inability to wean from the device within days of device placement, the possibility of transplant or permanent LVAD placement must be considered. Guidelines for use of ECMO are published by the Extracorporeal Life Support Organization and appropriately stress the need for a reversible condition to be present for utilization of ECMO to be most appropriate.⁵² Contraindications and complications are listed in Box 11.6.

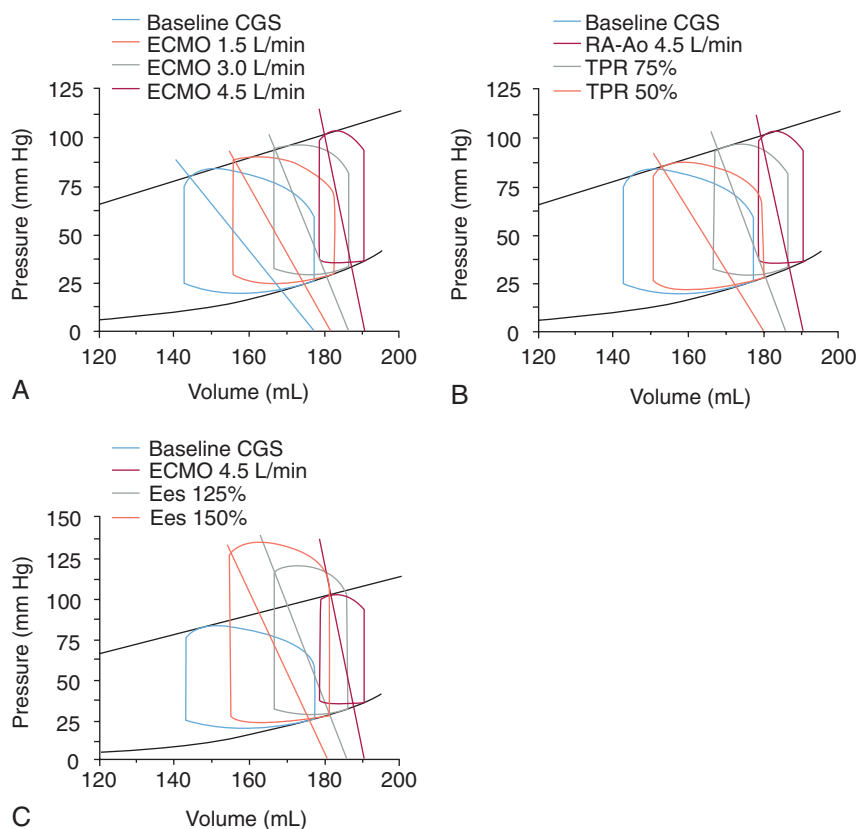


Fig. 11.40 Pressure-volume loops generated by the extracorporeal membrane oxygenator device. (A) Pressure-volume loops demonstrating increasing levels of ventricular support. Note the increase in myocardial oxygen demand as support levels increase. (B) The addition of afterload reduction and/or (C) increases in contractility improves the hemodynamics of the device. *CGS*, Cardiogenic shock; *ECMO*, extracorporeal membrane oxygenation; *Ees*, end-systolic elastance; *RA-Ao*, right atrial to aorta; *TPR*, total peripheral resistance. (From Burkhoff D, Sayer G, Doshi D, Uriel N. Hemodynamics of mechanical circulatory support. *J Am Coll Cardiol*. 2015;66(23):2663–2674.)

Box 11.6 Venoarterial Extracorporeal Membrane Oxygenator Synopsis

Venoarterial Extracorporeal Membrane Oxygenator

- **Insertion:** Femoral (percutaneous vs. cut down), central (right atrial to aorta), femoral venous-axillary
- Femoral venous cannula 23–25 mm
- Femoral aortic cannula 17–21 mm
- **How it works:** Large bore cannula drains blood from venous circulation; oxygenated blood is then returned to systemic arterial circulation
- Provides full cardiopulmonary support
- Good for biventricular failure or pulmonary compromise
- Generally requires deep sedation of patient
- **Contraindications:** Inability to anticoagulate, preexisting conditions incompatible with survival
- **Complications:** Bleeding, peripheral ischemia, hemolysis, vascular injury, transient ischemic attack/stroke, intracardiac thrombus (depending on access site), potential for incomplete cardiac drainage, poor cerebral oxygenation

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